The Real Number System

REAL NUMBERS

R = all numbers

Decimals that never end and have no repeating pattern. *Think ***MUMBO JUMBO NUMBERS***

IRRATIONAL NUMBERS

I = π, $\sqrt{2}$, $-\sqrt{101}$

All fractions and decimals that end or have a repeated pattern.

RATIONAL NUMBERS

Q = $\frac{1}{3}$, -0.67, $\sqrt{16}$

INTEGERS

Z = $\ldots, -2, -1, 0, 1, 2, \ldots$

WHOLE NUMBERS

W = $0, 1, 2, 3, \ldots$

NATURAL NUMBERS

N = $1, 2, 3, \ldots$

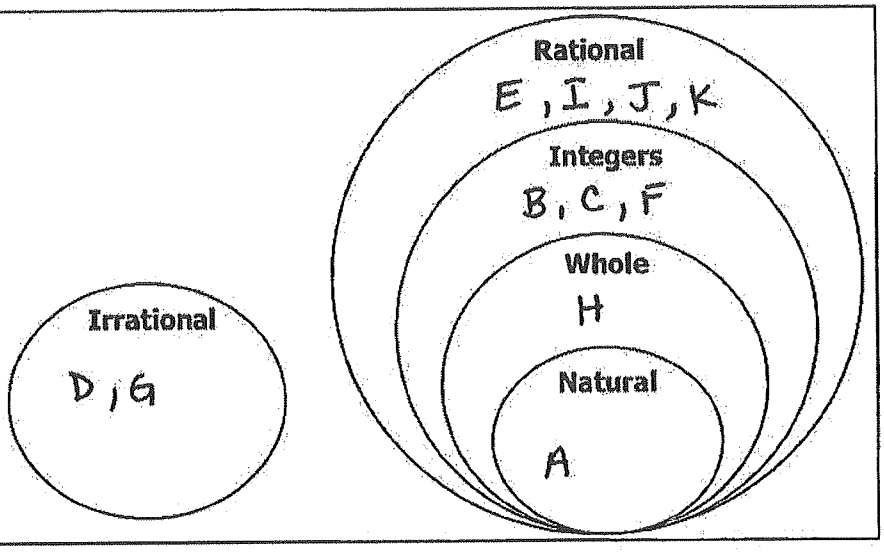
Directions: Name all sets of numbers to which each number belongs.		
1. 30 N, W, Z, Q, R	2. -11 Z, Q, R	3. $5\frac{4}{7}$ Q, R
4. $\sqrt{21}$ I, R	5. 0 W, Z, Q, R	6. $-\sqrt{9}$ Z, Q, R
7. $\frac{6}{3}$ N, W, Z, Q, R	8. π I, R	9. $5.\overline{3}$ Q, R
10. $\frac{\sqrt{144}}{-3}$ Z, Q, R	11. -2.875 Q, R	12. $\frac{\sqrt{17}}{\sqrt{17}}$ N, W, Z, Q, R

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Organizing the Real Numbers

Directions: Place the LETTER of each value in its location in the real number system below.

A. 7	B. $-\frac{36}{12}$
C. $-\sqrt{36}$	D. π
E. 6.48	F. -2
G. $-\sqrt{10}$	H. 0
I. $2\frac{1}{6}$	J. $-0.\overline{7}$
K. $\sqrt{\frac{4}{81}}$	



1. Which expression does **not** name an integer?

   A. -15 C. 0

   (B.) $\frac{6}{14}$ D. $\frac{12}{6}$

2. Which expression represents an irrational number?

   A. $0.\overline{18}$ C. $\frac{2}{3}$

   (B.) $\sqrt{75}$ D. $\sqrt{3}-\sqrt{3}$

3. Which number is **not** a whole number?

   A. 8 C. 0

   (B.) -10 D. $\frac{18}{3}$

4. Which of the following is a true statement?

   A. -9 is a whole number

   B. $\sqrt{25}$ is an irrational number

   C. 0 is a natural number

   (D.) $\frac{2}{3}$ is a rational number

5. Which of the following statements is false?

   (A.) All real numbers are rational numbers.

   B. Every integer is a rational number.

   C. All natural numbers are integers.

   D. Every whole number is a real number.

6. Give an example of a number that is a whole number, but not a natural number. 0

7. Give an example of a real number that is not rational. $\sqrt{3}, \pi$ * answers vary

8. Give an example of a rational number that is not an integer. 5.2 *

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 1:** The Real Numbers

Directions: Name all sets of numbers to which each real number belongs.

1. 12 N, W, Z, Q, R
2. -15 Z, Q, R
3. $1\frac{1}{2}$ Q, R
4. 3.18 Q, R
5. $\sqrt{48}$ I, R
6. $9.\overline{3}$ Q, R
7. $-2\frac{7}{9}$ Q, R
8. $\sqrt{25}$ N, W, Z, Q, R
9. $\sqrt{3}$ I, R
10. $-\sqrt{64}$ Z, Q, R
11. $-\sqrt{12}$ I, R
12. $\frac{8}{4}$ N, W, Z, Q, R

Directions: Place the LETTER of each value in its location in the real number system below.

A. 2.125 B. 0

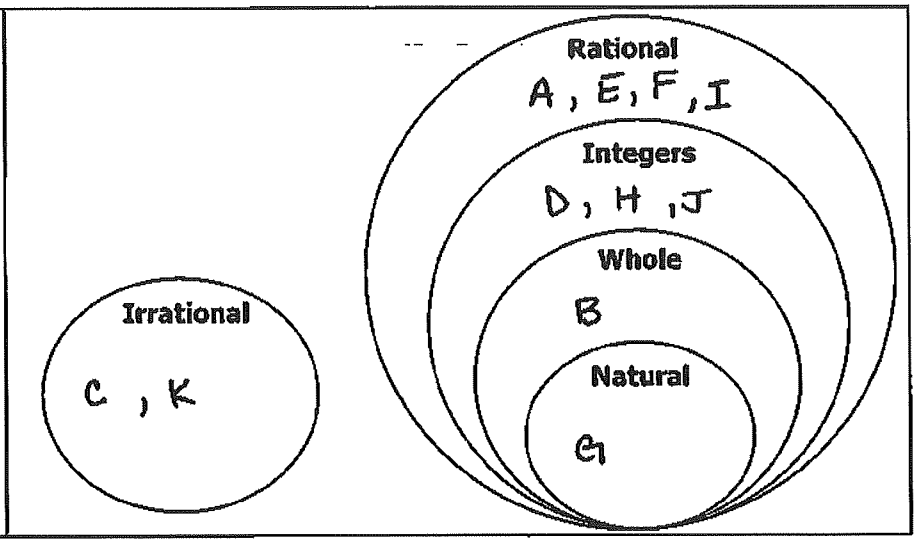
C. π D. $-\frac{15}{5}$

E. $3\frac{1}{4}$ F. $0.\overline{4}$

G. 23 H. -1

I. $\sqrt{\frac{16}{25}}$ J. $-\sqrt{49}$

K. $-\sqrt{35}$



Questions:

1. Give an example of a whole number that is not a natural number. 0
2. Give an example of a rational number that is not a whole number. 5/4 * answers vary
3. Give an example of a real number that is irrational. π *
4. Are integers always/sometimes/never whole numbers? Sometimes
5. Are natural numbers always/sometimes/never rational numbers? always
6. Are irrational numbers always/sometimes/never rational numbers? never

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Properties Warm-Up!!!

G 1. $16 = 16$

B 2. $7 + (a + b) = (7 + a) + b$

F 3. $(4y + 1) \cdot 0 = 0$

I 4. If $5 \cdot 2 = 10$ and $10 = \sqrt{100}$, then $5 \cdot 2 = \sqrt{100}$

C 5. $-2(x - y) = -2x + 2y$

E 6. $3 \cdot \frac{1}{3} = 1$

H 7. If $8x = 24$, then $24 = 8x$

A 8. $(2 + 1) + 9 = (1 + 2) + 9$

D 9. $6m \cdot 1 = 6m$

A. Commutative Property

B. Associative Property

C. Distributive Property

D. Identity Property

E. Inverse Property

F. Property of Zero

G. Reflexive Property

H. Symmetric Property

I. Transitive Property

10. **True/False:** Negative numbers closed under subtraction. Provide an example if false.

False; $-3 - (-7) = 4$ (not negative)

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
Absolute Value	The distance from zero on a number line

Examples

Directions: Evaluate each expression.

1. $\lvert -13 \rvert =$ 13	2. $\lvert 21 \rvert =$ 21	3. $\lvert -3 \rvert + \lvert -5 \rvert =$ $3+5=8$
4. $\lvert 9 \rvert + \lvert -8 \rvert =$ $9+8=17$	5. $\lvert -12 \rvert + \lvert 15 \rvert =$ $12+15=27$	6. $\lvert 21-17 \rvert =$ $\lvert 4 \rvert = 4$
7. $\lvert -11 \rvert - \lvert -5 \rvert =$ $11-5=6$	8. $\lvert 4 \rvert - \lvert -4 \rvert =$ $4-4=0$	9. $\lvert 15-23 \rvert =$ $\lvert -8 \rvert = 8$
10. $\lvert 24 \rvert - \lvert -15 \rvert =$ $24-15=9$	11. $\lvert 28 \rvert - \lvert -26 \rvert =$ $28-26=2$	12. $\lvert -3 \rvert - \lvert 19 \rvert =$ $3-19=-16$

Order of Operations

Simplifying an expression using a specific order

P	Parenthesis + Grouping Symbols
E	Exponents
M/D	Multiplication/Division from left → right
A/S	Addition/Subtraction from left → right

Examples

Directions: Evaluate each expression.

13. $7+54\div3(2)$

$=7+18(2)$

$=7+36$

$=43$

14. $17-2^5+4+6$

$=17-32\div4+6$

$=17-8+6$

$=9+6$

$=15$

15. $24-4^2\cdot3+15$

$=24-16\cdot3+15$

$=24-48+15$

$=-24+15$

$=-9$

16. $\dfrac{(7-2^2)+17}{-14+2\cdot5}$

$=\dfrac{(7-4)+17}{-14+10}$

$=\dfrac{20}{-4}=-5$

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17. $(8-5)^3-|5^2-4^3|\div 3$

$= (3)^3 - |25-64| \div 3$

$= 27 - |-39| \div 3$

$= 27 - 13$

$= \boxed{14}$

18. $|-7^2+3^2\cdot 4|+18\div 2\cdot 5$

$= |-49+9\cdot 4| + 9\cdot 5$

$= |-49+36| + 45$

$= |-13| + 45$

$= 13 + 45 = \boxed{58}$

19. $\dfrac{5^2-3^4}{(2^3-5)\cdot(6^2\div 9)}$

$= \dfrac{25-81}{(8-5)\cdot(36\div 9)}$

$= \dfrac{-56}{3\cdot 4} = \dfrac{-56}{12} = \boxed{\dfrac{-14}{3}}$

20. $-|-9-10|+12^2\div 6\cdot 3$

$= -|-19| + 144 \div 6\cdot 3$

$= -19 + 24\cdot 3$

$= -19 + 72$

$= \boxed{53}$

21. $20^2\div\{3(5-9)^2+2\}$

$= 400 \div \{3(-4)^2+2\}$

$= 400 \div \{3(16)+2\}$

$= 400 \div \{48+2\}$

$= 400 \div 50$

$= \boxed{8}$

22. $\dfrac{12[30-(9+4^2)]}{|10|-|-6|}$

$= \dfrac{12[30-(9+16)]}{10-6}$

$= \dfrac{12[30-25]}{4}$

$= \dfrac{12(5)}{4} = \dfrac{60}{4} = \boxed{15}$

23. $\dfrac{84}{12}+(5-7^2)-72\div 6\cdot 2$

$= 7+(5-49)-12\cdot 2$

$= 7+(-44)-24$

$= -37-24$

$= \boxed{-61}$

24. $\dfrac{-(8-11)^3-3|-14|}{2^4-7}$

$= \dfrac{-(-3)^3-3(14)}{16-7}$

$= \dfrac{-(-27)-42}{9}$

$= \dfrac{27-42}{9} = \dfrac{-15}{9} = \boxed{\dfrac{-5}{3}}$

Summary: _____

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Name: _____ Unit 1: Algebra Basics

Date: _____ Bell: _____ Homework 4: Order of Operations

Directions: Evaluate each expression using the order of operations.

1. $17-5\cdot4\div2$

$17-20\div2$

$17-10 = \boxed{7}$

2. $40-32\div8+5\cdot-2$

$40-4+(-10)$

$36-10 = \boxed{26}$

3. $35-14\div2+8^2$

$35-7+64$

$28+64 = \boxed{92}$

4. $(4-7)^2-6\cdot7+20$

$(-3)^2-42+20$

$9-42+20$

$-33+20 = \boxed{-13}$

5. $2[18-(5+3^2)+7]$

$2[18-(5+9)\div7]$

$2[18-14\div7]$

$2[18-2] = 2(16) = \boxed{32}$

6. $[(-5+1)\div2]^3-|-7|$

$[(-4)\div2]^3-7$

$[-2]^3-7$

$-8-7 = \boxed{-15}$

7. $1+(-2-5)^2+(14-17)\cdot4$

$1+(-7)^2+(-3)\cdot4$

$1+49+(-12)$

$50-12 = \boxed{38}$

8. $\dfrac{6(2+4)-1}{2\cdot3+1}$

$\dfrac{6(6)-1}{6+1} = \dfrac{36-1}{7} = \dfrac{35}{7}$

$= \boxed{5}$

9. $\dfrac{2^5\div8+3}{4+3}$

$\dfrac{32\div8+3}{7} = \dfrac{4+3}{7} = \dfrac{7}{7}$

$= \boxed{1}$

10. $\dfrac{17\cdot5-3\cdot5}{3^2+1}$

$\dfrac{85-15}{9+1} = \dfrac{70}{10} = \boxed{7}$

11. $\dfrac{13+7^2\div7}{9-20\div4-16}$

$\dfrac{13+49\div7}{9-5-16} = \dfrac{13+7}{4-16} = \dfrac{20}{-12}$

$= \boxed{-\dfrac{5}{3}}$

12. $-|-16|\div\dfrac{(3^2-4)}{5-10}$

$-16\div\dfrac{9-4}{-5} = -16\div\dfrac{5}{-5}$

$= -16\div-1$

$= \boxed{16}$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
ALGEBRAIC EXPRESSION	A mathematical phrase made up of numbers and variables.
SUBSTITUTION *Property*	If $a = b$, then a may be replaced for b in any expression
EVALUATING *Expressions*	**To evaluate an expression variable replacements:** • Substitute the variables with their given values. • Each time you substitute a variable with a number, put parenthesis around the number! • Follow the order of operations to evaluate!
EXAMPLES	**Directions:** Evaluate each expression using the variable replacements.

1. $ab^2 + c$ if $a = 2, b = 4, c = 7$

$(2)(4)^2 + 7$

$2(16) + 7$

$32 + 7 = \boxed{39}$

2. $3x^2 - 4x$ if $x = -2$

$3(-2)^2 - 4(-2)$

$3(4) + 8$

$12 + 8 = \boxed{20}$

YOU TRY!

Directions: Evaluate each expression using the variable replacements.

3. $a^2b - b^2$ if $a = 3$ and $b = -4$

$(3)^2(-4) - (-4)^2$

$9(-4) - (16)$

$-36 - 16 = \boxed{-52}$

4. $a^2b - b^2$ if $a = 4$ and $b = -7$

$(4)^2(-7) - (-7)^2$

$16(-7) - 49$

$-112 - 49 = \boxed{-161}$

5. $-y^2 - 3xy$ if $x = -4$ and $y = 2$

$-(2)^2 - 3(-4)(2)$

$-4 + 12(2)$

$-4 + 24 = \boxed{20}$

6. $-y^2 - 3xy$ if $x = -\frac{5}{6}$ and $y = -12$

$-(-12)^2 - 3(-5/6)(-12)$

$-144 + (5/2)(-12)$

$-144 - 30 = \boxed{-174}$

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7. $\frac{a^2+b^2}{a-b}$ if $a = 3$ and $b = 4$

$\frac{(3)^2+(4)^2}{3-4}$

$\frac{9+16}{-1} = \frac{25}{-1} = \boxed{-25}$

8. $\frac{a^2+b^2}{a-b}$ if $a = -3$ and $b = -5$

$\frac{(-3)^2+(-5)^2}{(-3)-(-5)}$

$\frac{9+25}{-3+5} = \frac{34}{2} = \boxed{17}$

9. $x^2-2(x-y)-z^3$ if $x = 2$, $y = -2$, and $z = -3$

$(2)^2-2(2-(-2))-(-3)^3$

$4-2(4)-(-27)$

$4-8+27$

$-4+27 = \boxed{23}$

10. $x^2-2(x-y)-z^3$ if $x = -3$, $y = 4$ and $z = -1$

$(-3)^2-2(-3-4)-(-1)^3$

$9-2(-7)-(-1)$

$9+14+1$

$23+1 = \boxed{24}$

11. $-a^2-2bc-|c|$ if $a = -2$, $b = 3$, and $c = -3$

$-(-2)^2-2(3)(-3)-|-3|$

$-4-6(-3)-3$

$-4+18-3$

$14-3 = \boxed{11}$

12. $-a^2-2bc-|c|$ if $a = -3$, $b = -5$, and $c = 2$

$-(-3)^2-2(-5)(2)-|2|$

$-9+10(2)-2$

$-9+20-2$

$11-2 = \boxed{9}$

Summary: _____

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Name: _____ Unit 1: Algebra Basics

Date: _____ Bell: _____ Homework 5: Evaluating Expressions

Directions: Evaluate each expression for the given variable values.

1. $n^2 - 3n + 8$ if $n = 4$

$(4)^2 - 3(4) + 8$

$16 - 12 + 8$

$4 + 8 = \boxed{12}$

2. $\frac{x^2 - 4y}{2}$ if $x = 4$ and $y = -3$

$\frac{(4)^2 - 4(-3)}{2} = \frac{16 + 12}{2} = \frac{28}{2}$

$= \boxed{14}$

3. $4|m - n|$ if $m = -7$ and $n = 2$

$4|(-7) - 2|$

$4|-9|$

$4(9) = \boxed{36}$

4. $19 - x^2$ if $x = -5$

$19 - (-5)^2$

$19 - 25 = \boxed{-6}$

5. $s^2t - 10$ if $s = -8$ and $t = \frac{3}{4}$

$(-8)^2(3/4) - 10$

$64(3/4) - 10$

$48 - 10 = \boxed{38}$

6. $4p^2 + 7q^3$ if $p = -3$ and $q = -2$

$4(-3)^2 + 7(-2)^3$

$4(9) + 7(-8)$

$36 - 56 = \boxed{-20}$

7. $\frac{(a+b)^5}{a - 2b^2}$ if $a = 6$ and $b = -8$

$\frac{(6 + (-8))^5}{6 - 2(-8)^2} = \frac{(-2)^5}{6 - 2(64)} = \frac{-32}{6 - 128}$

$= \frac{-32}{-122} = \boxed{\frac{16}{61}}$

8. $\frac{5c^2 - d^2 + 3}{2c - 4d}$ if $c = -1$ and $d = -4$

$\frac{5(-1)^2 - (-4)^2 + 3}{2(-1) - 4(-4)} = \frac{5 - 16 + 3}{-2 + 16}$

$= \frac{-8}{14} = \boxed{-\frac{4}{7}}$

9. $-q^2 - r^2 + 3s$ if $q = 9$, $r = -6$, $s = -20$

$-(9)^2 - (-6)^2 + 3(-20)$

$-81 - 36 - 60$

$-117 - 60 = \boxed{-177}$

10. $3|x + y|^2 - (xy)^2$ if $x = 3$ and $y = -5$

$3|3 + (-5)|^2 - (3 \cdot -5)^2$

$3|-2|^2 - (-15)^2$

$3(4) - 225$

$12 - 225 = \boxed{-213}$

11. $2x^2 - 5xy - y^3$ if $x = -3$ and $y = -2$

$2(-3)^2 - 5(-3)(-2) - (-2)^3$

$2(9) + 15(-2) - (-8)$

$18 - 30 + 8$

$-12 + 8 = \boxed{-4}$

12. $-a^2 + 7b^4 - 2c^3$ if $a = -4$, $b = -2$, $c = -3$

$-(-4)^2 + 7(-2)^4 - 2(-3)^3$

$-16 + 7(16) - 2(-27)$

$-16 + 112 + 54$

$96 + 54 = \boxed{150}$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples

Parts of an Expression

$7x + 3 - 5x + 8 + 4x - 1$

Variable Terms
(Terms WITH a variable)
$7x, -5x, 4x$

Constant Terms
(Terms WITHOUT a variable)
$3, 8, -1$

Coefficients
(Number NEXT TO a variable)
$7, -5, 4$

Combining Like Terms

You can simplify an algebraic expression by **combining like terms.** This means to combine common variable terms and contant terms.

Example: Simplify the expression below:

$7x + 3 - 5x + 8 + 4x - 1 = 6x + 10$

Examples

Directions: Simplify each expression.

1. $3x + 6x$ $9x$	**2.** $9a - 10a$ $-a$
3. $7m - 5m - 6$ $2m - 6$	**4.** $9 - 6w + 5$ $-6w + 14$
5. $7p - 1 - 9p + 5$ $-2p + 4$	**6.** $5h - 6 - 8 + 7h$ $12h - 14$
7. $5x + 7y - x + 2y + 3x$ $7x + 9y$	**8.** $26a + 2b - 4a + 10b$ $22a + 12b$

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with Exponents

Terms with the same exponents or combination of variables and exponents are considered like terms. When combined, be sure to NOT change their exponents!

Examples

9. $5x^2 + 7x^2$

$12x^2$

10. $-15n^5 + 4n^5$

$-11n^5$

11. $8c^3 - 7c^3 + 16c^3$

$17c^3$

12. $8k^2 - k - 5k + 7 - 2k^2$

$6k^2 - 6k + 7$

13. $-4a^2b + 7 + 10a^2b$

$6a^2b + 7$

14. $x^2 + 2xy - 7xy + 4y^2$

$x^2 - 5xy + 4y^2$

15. $a^2 + 7ab - 10ab + b^2$

$a^2 - 3ab + b^2$

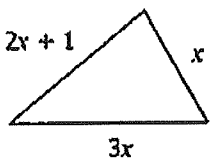
16. $4x^2 - 7 + 2x - 3x^2 + 8x - 6$

$x^2 + 10x - 13$

Geometric Applications

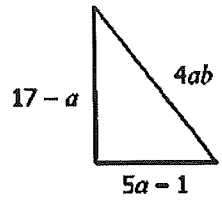
Directions: Give the perimeter of each figure as a simplified expression.

17.



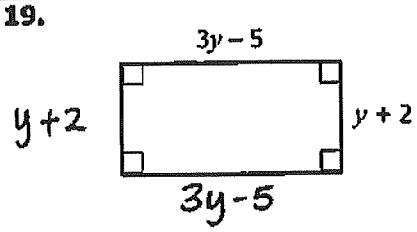
$6x + 1$

18.



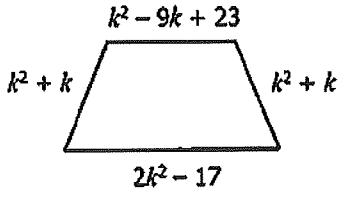
$4a + 4ab + 16$

19.



$8y - 6$

20.



$5k^2 - 7k + 6$

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ Bell: _____ **Homework 7:** Combining Like Terms

Directions: Simplify each expression by combining like terms.

1. $8y - 9y$

$-y$

2. $3x - 7x + 1$

$-4x+1$

3. $8a - 6 + a - 1$

$9a-7$

4. $-2y + 14 - 9y + 5$

$-11y+19$

5. $-x - 2 + 15x$

$14x-2$

6. $-7w - 5 - 7w + 9$

$-14w+4$

7. $8d - 4 - d - 2$

$7d-6$

8. $40 - 9x + 4x + 5$

$-5x+45$

9. $9a + 8 - 2a - 3 - 5a$

$2a+5$

10. $4xy - 6xy + 5y - 9x$

$-2xy+5y-9x$

11. $7x + 2y + 5 - 5x + 9y - 1$

$2x+11y+4$

12. $-5f - 7g + f - 5g - 4 + f$

$-3f-12g-4$

13. $-6 + 3a - b - 4a + 4b + 5$

$-a+3b-1$

14. $6n^2 - 2n - 3 + 5n^2 - 9n - 6$

$11n^2-11n-9$

15. $8c^2 - c + 3 + 2c^2 - c + 2$

$10c^2-2c+5$

16. $-12x^2y - 3x^2y$

$-15x^2y$

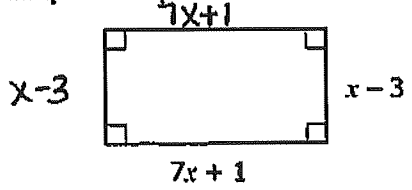
17. $6a^2b^2 + 3ab^2 - a^2b^2 - 4ab^2$

$5a^2b^2-ab^2$

18. $19x^2y - 4xy^2 + 5xy^2 - 16x^2y$

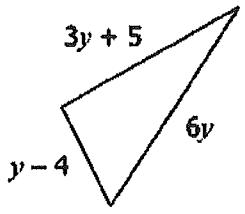
$3x^2y+xy^2$

19. Write the perimeter of the rectangle as a simplified expression.



$16x-4$

20. Write the perimeter of the triangle as a simplified expression.



$10y+1$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples		
Distributive Property Recall: $a(b+c)=$ $ab+ac$ or $a(b-c)=$ $ab-ac$	1. $7(x+4)$ $7x+28$	2. $2(b-3)$ $2b-6$	3. $-4(y+3)$ $-4y-12$
	4. $-5(m-2)$ $-5m+10$	5. $-(y-9)$ $-y+9$	6. $8(4-b)$ $32-8b$
	7. $-4(-w-10)$ $4w+40$	8. $5(2m-3)$ $10m-15$	9. $-3(2x-4)$ $-6x+12$
	10. $3(m+n)$ $3m+3n$	11. $x(y+4)$ $xy+4x$	12. $d(c-4)$ $dc-4d$
	13. $6a(b+2c)$ $6ab+12ac$	14. $2(x-y+8)$ $2x-2y+16$	15. $x(2y-5z+1)$ $2xy-5xz+x$
	16. $\frac{1}{4}(8x-4)$ $2x-1$	17. $\frac{1}{6}\left(\frac{3}{5}x+18\right)$ $\frac{1}{10}x+3$	18. $-\frac{1}{8}\left(\frac{4}{5}x-24\right)$ $-\frac{1}{10}x+3$
Distribute AND Combine!	To **simplify an expression** means to ensure there are **no parentheses** and **no like terms.** In order to do this, distribute first (if needed), then combine like terms.		
Examples	19. $8(2x-3)-6x$ $16x-24-6x$ $=\boxed{10x-24}$	20. $9(2k-4)-2(7k-12)$ $18k-36-14k+24$ $=\boxed{4k-12}$	
	21. $10-(y-6)-y$ $10-y+6-y$ $=\boxed{-2y+16}$	22. $6+8(4w-7)-(2w+1)$ $6+32w-56-2w-1$ $=\boxed{30w-51}$	
	23. $\frac{9}{4}\left(\frac{10}{3}x-4\right)-\frac{1}{4}(2x-8)$ $\frac{15}{2}x-9-\frac{1}{2}x+2$ $=\boxed{7x-7}$	24. $\frac{7}{8}\left(\frac{4}{5}x+16\right)+\frac{3}{5}\left(\frac{1}{2}x-10\right)$ $\frac{7}{10}x+14+\frac{3}{10}x-6$ $=\boxed{x+8}$	

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 8:** Simplifying Expressions

Directions: Simplify each expression by distributing.	
1. $8(x + 5) =$ $8x + 40$	2. $3(2y - 7) =$ $6y - 21$
3. $-2(3m + 9) =$ $-6m - 18$	4. $-6(z - 1) =$ $-6z + 6$
5. $-(5x + 1) =$ $-5x - 1$	6. $-(x + 4) =$ $-x - 4$
7. $b(a - 6) =$ $ab - 6b$	8. $7a(3b - 2c + 4) =$ $21ab - 14ac + 28a$
9. $\frac{1}{2}(8x + 2) =$ $4x + 1$	10. $\frac{2}{3}\left(\frac{1}{4}x - 6\right) =$ $\frac{1}{6}x - 4$

Directions: Simplify each expression by distributing and combining like terms	
11. $-7(w - 4) + 3w - 27$ $-7w + 28 + 3w - 27$ $= \boxed{-4x + 1}$	12. $6(x + 1) - 5(x + 2)$ $6x + 6 - 5x - 10$ $= \boxed{x - 4}$
13. $-5(k + 6) + 7(k - 4)$ $-5k - 30 + 7k - 28$ $= \boxed{2k - 58}$	14. $8(x + 5) - 4(x + 4)$ $8x + 40 - 4x - 16$ $= \boxed{4x + 24}$
15. $-9(m + 2) + 4(6 - 7m)$ $-9m - 18 + 24 - 28m$ $= \boxed{-37m + 6}$	16. $-3(-9v - 4) - 2(v - 2)$ $27v + 12 - 2v + 4$ $= \boxed{25v + 16}$
17. $-4(8 - c) - (3c + 7)$ $-32 + 4c - 3c - 7$ $= \boxed{c - 39}$	18. $6(p + 6) - (p - 6)$ $6p + 36 - p + 6$ $= \boxed{5p + 42}$
19. $21 - 8(w + 3) + 3 + 7w$ $21 - 8w - 24 + 3 + 7w$ $= \boxed{-w}$	20. $\frac{1}{2}\left(\frac{3}{5}x - 8\right) + \frac{7}{4}\left(\frac{2}{5}x + 4\right)$ $\frac{3}{10}x - 4 + \frac{7}{10}x + 7 = \boxed{x + 3}$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples

Key Words & Phrases

ADDITION +

- sum
- total
- plus
- increased by
- more than

SUBTRACTION −

- difference
- minus
- decreased by
- less than *
- subtracted from *

MULTIPLICATION ●

- times
- product
- of
- twice
- double / triple

DIVISON /

- quotient
- separate
- divided by
- half / third
-

(Parentheses Phrases)

- times "the sum of"
- times "the difference of / between"

"Turn Around" Phrases

- less than
- subtracted from

Examples

Directions: Translate each expression or equation.

1. "the product of a number and 3"
   $3n$
2. "the difference between a number and 10"
   $n-10$
3. "the quotient of a number and -5"
   $\frac{n}{-5}$
4. "four subtracted from a number"
   $n-4$
5. "the sum of twice a number and 7"
   $2n+7$
6. "the product of a number and 10, increased by 1"
   $10n+1$

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	7. "four less than a number squared" n^2-4	**8.** "twice a number divided by 5" $\frac{2n}{5}$
Equations	**9.** "the sum of twice a number and 10 is 36" $2n+10=36$	**10.** "three times the difference of a number and 7 is 18" $3(n-7)=18$
	11. "four times a number divided by 8 is 3" $\frac{4n}{8}=3$	**12.** "five times the sum of a number and 6 is 48" $5(n+6)=48$
	13. "one half of a number is 8 less than the number itself" $\frac{1}{2}n = n-8$	**14.** "the product of a number and 3 is the same as the sum of that number and six" $3n = n+6$
	15. "fifteen decreased by the quotient of a number and 3 is 20" $15-\frac{n}{3}=20$	**16.** "four less than the product of a number and 7 is eight more than that number" $7n-4 = 8+n$

Inequalities	**LESS THAN**	**LESS THAN OR EQUAL TO**	**GREATER THAN**	**GREATER THAN OR EQUAL TO**
	$<$	$\leq$	$>$	$\geq$

Examples		
	17. "a number is less than -10" $n<-10$	**18.** "a number is less than or equal to 8" $n\leq 8$
	19. "a number is greater than or equal to -4" $n\geq -4$	**20.** "a number is at most 45" $n\leq 45$
	21. "a speed limit of 55 mph" $n\leq 55$	**22.** "a number is at least 16" $n\geq 16$
	23. "a number is no less than 10" $n\geq 10$	**24.** "a number is no more than 26" $n\leq 26$
	25. "a maximum number of 25" $n\leq 25$	**26.** "a minimum number of 5" $n\geq 5$

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 9:** Translating Expressions Equations, & Inequalities

Directions: Translate each expression.	
1. "the sum of a number and 4." $n+4$	**2.** "the quotient of a number and -2." $\frac{n}{-2}$
3. "the product of a number and 5." $5n$	**4.** "the difference of a number and 7." $n-7$
5. "a number squared increased by 1" n^2+1	**6.** "three less than twice a number" $2n-3$

Directions: Translate each equation. ***DO NOT SOLVE!***	
7. "twice a number divided by 6 is 42." $\frac{2n}{6}=42$	**8.** "two times the sum of number and 5 is 20" $2(n+5)=20$
9. "four times the difference of a number and 7 is 32" $4(n-7)=32$	**10.** "one-third of a number is 12 less than the number itself" $\frac{1}{3}n = n-12$
11. "ten increased by the quotient of a number and 2 is -15" $10+\frac{n}{2}=-15$	**12.** "one less than the product of a number and 3 is 5 more than the number itself" $3n-1=5+n$

Directions: Translate each inequality.	
13. "a number less than 15" $n<15$	**14.** "a number greater than or equal to -3" $n\geq -3$
15. "a number that is at least 90" $n\geq 90$	**16.** "a speed limit of 65 mph" $n\leq 65$
17. "a number that is no more than 20" $n\leq 20$	**18.** "a maximum number of 50" $n\leq 50$
19. "a minimum number of 10" $n\geq 10$	**20.** "a number no less than 25" $n\geq 25$

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WARM-UP!!!

Translate the following expressions, equations, & inequalities:

1. "The product of a number and 7" — $7n$
2. "Nine subtracted from twice a number" — $2n - 9$
3. "One less than the quotient of a number and -5." — $\frac{n}{-5} - 1$
4. "Three times the sum of a number and 10 is 24" — $3(n + 10) = 24$
5. "The difference of twice a number and 3 is -21." — $2n - 3 = -21$
6. "One-third of a number increased by 8 is 37." — $\frac{1}{3}n + 8 = 37$
7. "x is at most 6" — $x \leq 6$
8. "You must be at least 18 years old to vote" — $x \geq 18$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
One-Step Equations	1. $m + 12 = 10$ $-12 \quad -12$ $m = -2$	2. $-2 = g - 9$ $+9 \quad +9$ $7 = g$
	3. $\frac{-7y}{-7} = \frac{-91}{-7}$ $y = 13$	4. $\frac{a}{9} = -4 \cdot 9$ $a = -36$
Fractions To "get rid" of a fraction, multiply by the reciprocal!	5. $\frac{3}{2} \cdot \frac{2}{3}x = 10 \cdot \frac{3}{2}$ $x = 15$	6. $\frac{9}{4} \cdot \frac{4}{9}w = -8 \cdot \frac{9}{4}$ $w = -18$
	7. $-\frac{5}{6} \cdot -\frac{6}{5}k = 12 \cdot \frac{-5}{6}$ $k = -10$	8. $-2 \cdot -\frac{1}{2}m = -9 \cdot -2$ $m = 18$
Two-Step Equations	**To Solve a Two-Step Equation:** 1. Undo the **Addition/Subtraction** (to remove constant term) 2. Undo the **Multiplication/Division** (to remove coefficient)	
	9. $6x + 8 = 50$ $-8 \quad -8$ $\frac{6x}{6} = \frac{42}{6}$ $x = 7$	10. $2n - 5 = 11$ $+5 \quad +5$ $\frac{2n}{2} = \frac{16}{2}$ $n = 8$
	11. $13 = -4k + 9$ $-9 \quad -9$ $\frac{4}{-4} = \frac{-4k}{-4}$ $k = -1$	12. $7 - 3y = 34$ $-7 \quad -7$ $\frac{-3y}{-3} = \frac{27}{-3}$ $y = -9$

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13. $\frac{x}{2} - 7 = 9$

$+7 \quad +7$

$2 \cdot \frac{x}{2} = 16 \cdot 2$

$\boxed{x = 32}$

14. $11 = \frac{c}{-5} + 8$

$-8 \quad -8$

$-5 \cdot 3 = \frac{c}{-5} \cdot -5$

$\boxed{-15 = c}$

15. $\frac{3}{5}x + 22 = 28$

$-22 \quad -22$

$\frac{5}{3} \cdot \frac{3}{5}x = 6 \cdot \frac{5}{3}$

$\boxed{x = 10}$

16. $-\frac{1}{3}m + 1 = -7$

$-1 \quad -1$

$-3 \cdot -\frac{1}{3}m = -8 \cdot -3$

$\boxed{m = 24}$

17. $-10 \div \frac{7}{4}p = -38$

$+10 \quad +10$

$\frac{4}{7} \cdot \frac{7}{4}p = -28 \cdot \frac{4}{7}$

$\boxed{p = -16}$

18. $15 = 9 - \frac{1}{2}x$

$-9 \quad -9$

$-2 \cdot 6 = -\frac{1}{2}x \cdot -2$

$\boxed{-12 = x}$

Watch Out!

The examples below are different in that the multiplication/division is done FIRST, followed by the addition/subtraction.

19. $\frac{x+11}{8} = -3 \cdot 8$

$x + 11 = -24$

$-11 \quad -11$

$\boxed{x = -35}$

20. $\frac{n-5}{-2} = -7 \cdot -2$

$n - 5 = 14$

$+5 \quad +5$

$\boxed{n = 19}$

21. $1 = \frac{a-13}{-6} \cdot -6$

$-6 = a - 13$

$+13 \quad +13$

$\boxed{7 = a}$

22. $4 = \frac{w+8}{9} \cdot 9$

$36 = w + 8$

$-8 \quad -8$

$\boxed{28 = w}$

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ Bell: _____ **Homework 10:** One & Two-Step Equations

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $y - 17 = -37$

$+17 \quad +17$

$y = -20$

2. $-12 + k = 18$

$+12 \quad +12$

$k = 30$

3. $5m = -20$

$\frac{5m}{5} = \frac{-20}{5}$

$m = -4$

4. $\frac{x}{9} = 3$ $\cdot 9$

$x = 27$

5. $\frac{4}{5}z = 16$ $\cdot \frac{5}{4}$

$z = 20$

6. $3p - 4 = 11$

$+4 \quad +4$

$\frac{3p}{3} = \frac{15}{3}$

$p = 5$

7. $-5a + 7 = -38$

$-7 \quad -7$

$\frac{-5a}{-5} = \frac{-45}{-5}$

$a = 9$

8. $31 - 12n = 211$

$-31 \quad -31$

$\frac{-12n}{-12} = \frac{180}{-12}$

$n = -15$

9. $\frac{x}{3} - 15 = -7$

$+15 \quad +15$

$3 \cdot \frac{x}{3} = 8 \cdot 3$

$x = 24$

10. $\frac{f}{-4} + 11 = 5$

$-11 \quad -11$

$-4 \cdot \frac{f}{-4} = -6 \cdot -4$

$f = 24$

11. $\frac{4}{3}v + 25 = 33$

$-25 \quad -25$

$\frac{3}{4} \cdot \frac{4}{3}v = 8 \cdot \frac{3}{4}$

$v = 6$

12. $\frac{5}{8}c + 13 = 18$

$-13 \quad -13$

$\frac{8}{5} \cdot \frac{5}{8}c = 5 \cdot \frac{8}{5}$

$c = 8$

13. $-\frac{2}{3}w + 53 = 59$

$-53 \quad -53$

$-\frac{3}{2} \cdot -\frac{2}{3}w = 6 \cdot -\frac{3}{2}$

$w = -9$

14. $11 = \frac{4}{5}h - 1$

$+1 \quad +1$

$\frac{5}{4} \cdot 12 = \frac{4}{5}h \cdot \frac{5}{4}$

$15 = h$

15. $-\frac{3}{8}r + 41 = -22$

$-41 \quad -41$

$-\frac{8}{3} \cdot -\frac{3}{8}r = -63 \cdot -\frac{8}{3}$

$r = 168$

16. $34 = 28 - \frac{2}{5}y$

$-28 \quad -28$

$-\frac{5}{2} \cdot 6 = -\frac{2}{5}y \cdot -\frac{5}{2}$

$-15 = y$

17. $\frac{n+4}{-2} = -9 \cdot -2$

$n + 4 = 18$

$-4 \quad -4$

$n = 14$

18. $-5 = \frac{x-1}{8} \cdot 8$

$-40 = x - 1$

$+1 \quad +1$

$-39 = x$

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Translate... and Solve!

1) "Twice a number plus 4 is 8"	$2n+4=8$ $-4\ \ -4$ $\frac{2n}{2}=\frac{4}{2}$ $\boxed{n=2}$
2) "The product of 4 and a number plus 17 is 5."	$4n+17=5$ $-17\ \ -17$ $\frac{4n}{4}=\frac{-12}{4}$ $\boxed{n=-3}$
3) "Twice a number minus 5 is 7"	$2n-5=7$ $+5\ \ +5$ $\frac{2n}{2}=\frac{12}{2}$ $\boxed{n=6}$
4) "The quotient of a number and 6 minus 3 is 1"	$\frac{n}{6}-3=1$ $+3\ \ +3$ $6\cdot\frac{n}{6}=4\cdot 6$ $\boxed{n=24}$
5) "Two more than 4 times a number is -18"	$4n+2=-18$ $-2\ \ -2$ $\frac{4n}{4}=\frac{-20}{4}$ $\boxed{n=-5}$
6) "Seven less than twice a number is 5"	$2n-7=5$ $+7\ \ +7$ $\frac{2n}{2}=\frac{12}{2}$ $\boxed{n=6}$
7) "Five less than the quotient of a number and 3 is -7"	$\frac{n}{3}-5=-7$ $+5\ \ +5$ $3\cdot\frac{n}{3}=-2\cdot 3$ $\boxed{n=-6}$
8) "One subtracted from the product of 4 and a number is 11"	$4n-1=11$ $+1\ \ +1$ $\frac{4n}{4}=\frac{12}{4}$ $\boxed{n=3}$
9) "The quotient of a number and -9, increased by 10 is 11"	$\frac{n}{-9}+10=11$ $-10\ \ -10$ $-9\cdot\frac{n}{-9}=1\cdot -9$ $\boxed{n=-9}$
10) "Half of a number decreased by 8 is -3"	$\frac{1}{2}n-8=-3$ $+8\ \ +8$ $2\cdot\frac{1}{2}n=5\cdot 2$ $\boxed{n=10}$

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11 "The sum of three-fourths of a number and 1 is -5"	$\frac{3}{4}n + 1 = -5$ $-1 \quad -1$ $\frac{4}{3}\cdot\frac{3}{4}n = -6 \cdot \frac{4}{3}$ $\boxed{n = -8}$	12 "Ten minus one-third of a number is 4"	$10 - \frac{1}{3}n = 4$ $-10 \quad -10$ $-3 \cdot -\frac{1}{3}n = -6 \cdot -3$ $\boxed{n = 18}$
13 "Four more than the quotient of a number and 3 is at least 9"	$4 + \frac{n}{3} \geq 9$ $-4 \quad -4$ $3 \cdot \frac{n}{3} \geq 5 \cdot 3$ $\boxed{n \geq 15}$	14 "Negative three times a number increased by seven is less than -11"	$-3n + 7 < -11$ $-7 \quad -7$ $\frac{-3n}{-3} < \frac{-18}{-3}$ $\boxed{n > 6}$
15 "Seven more than one-sixth of a number is greater than three"	$7 + \frac{1}{6}n > 3$ $-7 \quad -7$ $6 \cdot \frac{1}{6}n > -4 \cdot 6$ $\boxed{n > -24}$	16 "The sum of 5 and the twice a number is at most 27"	$5 + 2n \leq 27$ $-5 \quad -5$ $\frac{2n}{2} \leq \frac{22}{2}$ $\boxed{n \leq 11}$
17 "Four less than two-fifths of a number is no less than 2"	$\frac{2}{5}n - 4 \geq 2$ $+4 \quad +4$ $\frac{5}{2}\cdot\frac{2}{5}n \geq 6 \cdot \frac{5}{2}$ $\boxed{n \geq 15}$	18 "Three-fourths of a number decreased by 10 is greater than or equal 5"	$\frac{3}{4}n - 10 \geq 5$ $+10 \quad +10$ $\frac{4}{3}\cdot\frac{3}{4}n \geq 15 \cdot \frac{4}{3}$ $\boxed{n \geq 20}$
19 "The quotient of a number and -5, increased by 7 is at minimum 1"	$\frac{n}{-5} + 7 \geq 1$ $-7 \quad -7$ $-5 \cdot \frac{n}{-5} \geq -6 \cdot -5$ $\boxed{n \leq 30}$	20 "One less than half a number is a maximum of 19"	$\frac{1}{2}n - 1 \leq 19$ $+1 \quad +1$ $2\frac{1}{2}n \leq 20 \cdot 2$ $\boxed{n \leq 40}$

GOOD JOB!

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
Steps to Solve a Multi-Step Equation	1. Distribute (if necessary) 2. Combine like terms. 3. Solve!
Examples	

1. $9x + 1 - 7x - 5 = -20$

$2x - 4 = -20$

$+4 \quad +4$

$\frac{2x}{2} = \frac{-16}{2}$

$x = -8$

2. $91 = -7(3a - 1)$

$91 = -21a + 7$

$-7 \quad -7$

$\frac{84}{-21} = \frac{-21a}{-21}$

$a = -4$

3. $4m - 5(3m + 10) = 126$

$4m - 15m - 50 = 126$

$-11m - 50 = 126$

$+50 \quad +50$

$\frac{-11m}{-11} = \frac{176}{-11}$

$m = -16$

4. $-3(k - 8) - (k + 5) = 23$

$-3k + 24 - k - 5 = 23$

$-4k + 19 = 23$

$-19 \quad -19$

$\frac{-4k}{-4} = \frac{4}{-4}$

$k = -1$

5. $20 = 10x - 6(2x + 5)$

$20 = 10x - 12x - 30$

$20 = -2x - 30$

$+30 \quad +30$

$\frac{50}{-2} = \frac{-2x}{-2}$

$x = -25$

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6. $8(2w-1)-4w=-116$

$16w-8-4w=-116$

$12w-8=-116$

$+8 \quad +8$

$\frac{12w}{12}=\frac{-108}{12}$

$w=-9$

7. $11h-(2h-1)=118$

$11h-2h+1=118$

$9h+1=118$

$-1 \quad -1$

$\frac{9h}{9}=\frac{117}{9}$

$h=13$

8. $-25=\frac{1}{2}(10x-2)+3x$

$-25=5x-1+3x$

$-25=8x-1$

$+1 \quad +1$

$\frac{-24}{8}=\frac{8x}{8}$

$x=-3$

9. $7-\frac{5}{2}(8r-6)+2r=32$

$7-20r+15+2r=32$

$-18r+22=32$

$-22 \quad -22$

$\frac{-18r}{-18}=\frac{10}{-18}$

$r=-\frac{5}{9}$

Translate & Solve



10. *"Five times the difference of twice a number and three, decreased by the sum of the number and eight, equals 13."*

$5(2n-3)-(n+8)=13$

$10n-15-n-8=13$

$9n-23=13$

$+23 \quad +23$

$\frac{9n}{9}=\frac{36}{9}$

$n=4$

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Name: _____ **Unit 2:** Equations & Inequalities

Date: _____ Bell: _____ **Homework 1:** Multi-Step Equations

**** This is a 2-page document! ****

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $-4x - 7 - 3x + 4 = 25$

$-7x - 3 = 25$

$+3 \quad +3$

$\frac{-7x}{-7} = \frac{28}{-7}$

$\boxed{x = -4}$

2. $-4(7a + 5) = -160$

$-28a - 20 = -160$

$+20 \quad +20$

$\frac{-28a}{-28} = \frac{-140}{-28}$

$\boxed{a = 5}$

3. $3k + 2(5k - 3) = 7$

$3k + 10k - 6 = 7$

$13k - 6 = 7$

$+6 \quad +6$

$\frac{13k}{13} = \frac{13}{13}$

$\boxed{k = 1}$

4. $-79 = 7w + 3(4w - 1)$

$-79 = 7w + 12w - 3$

$-79 = 19w - 3$

$+3 \quad +3$

$\frac{-76}{19} = \frac{19w}{19}$

$\boxed{-4 = w}$

5. $9 - 4(2p - 1) = 41$

$9 - 8p + 4 = 41$

$-8p + 13 = 41$

$-13 \quad -13$

$\frac{-8p}{-8} = \frac{28}{-8}$

$\boxed{p = -\frac{7}{2}}$

6. $y - 3(2y - 7) = 76$

$y - 6y + 21 = 76$

$-5y + 21 = 76$

$-21 \quad -21$

$\frac{-5y}{-5} = \frac{55}{-5}$

$\boxed{y = -11}$

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7. $15 - 5(4c - 7) = 50$

$15 - 20c + 35 = 50$

$-20c + 50 = 50$

$-50 \quad -50$

$\frac{-20c}{-20} = \frac{0}{-20}$

$\boxed{c = 0}$

8. $-(6h + 7) + 8 = -19$

$-6h - 7 + 8 = -19$

$-6h + 1 = -19$

$-1 \quad -1$

$\frac{-6h}{-6} = \frac{-20}{-6}$

$\boxed{h = \frac{10}{3}}$

9. $-\frac{1}{3}(9x + 42) - 5x = -70$

$-3x - 14 - 5x = -70$

$-8x - 14 = -70$

$+14 \quad +14$

$\frac{-8x}{-8} = \frac{-56}{-8}$

$\boxed{x = 7}$

10. $8 = -18 + \frac{3}{8}(16 - 40n)$

$8 = -18 + 6 - 15n$

$8 = -12 - 15n$

$+12 \quad +12$

$\frac{20}{-15} = \frac{-15n}{-15}$

$\boxed{-\frac{4}{3} = n}$

11. $2(5v + 3) - 4(7v + 1) = -88$

$10v + 6 - 28v - 4 = -88$

$-18v + 2 = -88$

$-2 \quad -2$

$\frac{-18v}{-18} = \frac{-90}{-18}$

$\boxed{v = 5}$

12. $7(3r - 1) - (r + 5) = -52$

$21r - 7 - r - 5 = -52$

$20r - 12 = -52$

$+12 \quad +12$

$\frac{20r}{20} = \frac{-40}{20}$

$\boxed{r = -2}$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
Steps to Solve	1. Distribute (if necessary) 2. Combine like terms. 3. Move variables to one side. 4. Solve!

Examples

1. $5y - 8 = 3y + 12$

$-3y \quad -3y$

$2y - 8 = 12$

$+8 \quad +8$

$\frac{2y}{2} = \frac{20}{2}$

$\boxed{y = 10}$

2. $-6x + 14 = 12 - 8x$

$+8x \quad +8x$

$2x + 14 = 12$

$-14 \quad -14$

$\frac{2x}{2} = \frac{-2}{2}$

$\boxed{x = -1}$

3. $7k = 3k - 36$

$-3k \quad -3k$

$\frac{4k}{4} = \frac{-36}{4}$

$\boxed{k = -9}$

4. $12 - 2u = 9u + 45$

$+2u \quad +2u$

$12 = 11u + 45$

$-45 \quad -45$

$\frac{-33}{11} = \frac{11u}{11}$

$\boxed{-3 = u}$

5. $11 - m = 51 - 6m$

$+6m \quad +6m$

$11 + 5m = 51$

$-11 \quad -11$

$\frac{5m}{5} = \frac{40}{5}$

$\boxed{m = 8}$

6. $-10k + 1 = 40 - 7k$

$+10k \quad +10k$

$1 = 40 + 3k$

$-40 \quad -40$

$\frac{-39}{3} = \frac{3k}{3}$

$\boxed{-13 = k}$

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7. $3(6p - 1) = 11p - 45$

$18p - 3 = 11p - 45$

$-11p \quad -11p$

$7p - 3 = -45$

$+3 \quad +3$

$\frac{7p}{7} = \frac{-42}{7}$

$\boxed{p = -6}$

8. $2(4w - 1) = -10(w - 3) + 4$

$8w - 2 = -10w + 30 + 4$

$8w - 2 = -10w + 34$

$+10w \quad +10w$

$18w - 2 = 34$

$+2 \quad +2$

$\frac{18w}{18} = \frac{36}{18}$

$\boxed{w = 2}$

9. $9(m - 3) + 3m = 7m + 43$

$9m - 27 + 3m = 7m + 43$

$12m - 27 = 7m + 43$

$-7m \quad -7m$

$5m - 27 = 43$

$+27 \quad +27$

$\frac{5m}{5} = \frac{70}{5}$

$\boxed{m = 14}$

10. $-16 - 7(2a + 3) = 23 - 2a$

$-16 - 14a - 21 = 23 - 2a$

$-14a - 37 = 23 - 2a$

$+14a \quad +14a$

$-37 = 23 + 12a$

$-23 \quad -23$

$\frac{-60}{12} = \frac{12a}{12}$

$\boxed{-5 = a}$

11. $5x - (x - 18) = 6 - 2(x + 15)$

$5x - x + 18 = 6 - 2x - 30$

$4x + 18 = -2x - 24$

$+2x \quad +2x$

$6x + 18 = -24$

$-18 \quad -18$

$\frac{6x}{6} = \frac{-42}{6}$

$\boxed{x = -7}$

12. $8(y + 4) - 2(y - 1) = 70 - 3y$

$8y + 32 - 2y + 2 = 70 - 3y$

$6y + 34 = 70 - 3y$

$+3y \quad +3y$

$9y + 34 = 70$

$-34 \quad -34$

$\frac{9y}{9} = \frac{36}{9}$

$\boxed{y = 4}$

Summary: _____

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Name: _____ **Unit 2:** Equations & Inequalities

Date: _____ Bell: _____ **Homework 2:** Variables on Both Sides

**** This is a 2-page document! ****

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $4x + 1 = 2x - 5$

$-2x \quad -2x$

$2x + 1 = -5$

$-1 \quad -1$

$\frac{2x}{2} = \frac{-6}{2}$

$\boxed{x = -3}$

2. $y - 3 = -3y - 43$

$+3y \quad +3y$

$4y - 3 = -43$

$+3 \quad +3$

$\frac{4y}{4} = \frac{-40}{4}$

$\boxed{y = -10}$

3. $7 - 14m = 2m - 5$

$+14m \quad +14m$

$7 = 16m - 5$

$+5 \quad +5$

$\frac{12}{16} = \frac{16m}{16}$

$\boxed{\frac{3}{4} = m}$

4. $-10b + 5 = 7b + 5$

$+10b \quad +10b$

$5 = 17b + 5$

$-5 \quad -5$

$\frac{0}{17} = \frac{17b}{17}$

$\boxed{0 = b}$

5. $r + 15 = 4r - 6$

$-r \quad -r$

$15 = 3r - 6$

$+6 \quad +6$

$\frac{21}{3} = \frac{3r}{3}$

$\boxed{7 = r}$

6. $10 - 2v = -5v - 50$

$+5v \quad +5v$

$10 + 3v = -50$

$-10 \quad -10$

$\frac{3v}{3} = \frac{-60}{3}$

$\boxed{v = -20}$

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7. $3(5y + 1) = 18y$

$15y + 3 = 18y$

$-15y \quad -15y$

$3 = 3y$

$\frac{3}{3} \quad \frac{3y}{3}$

$\boxed{1 = y}$

8. $4 - 3x = 8 - 4x$

$+4x \quad +4x$

$4 + x = 8$

$-4 \quad -4$

$\boxed{x = 4}$

9. $8(2w - 2) = 7(3w + 2)$

$16w - 16 = 21w + 14$

$-16w \quad -16w$

$-16 = 5w + 14$

$-14 \quad -14$

$\frac{-30}{5} = \frac{5w}{5}$

$\boxed{-6 = w}$

10. $3(5x + 2) = 2(3x - 6)$

$15x + 6 = 6x - 12$

$-6x \quad -6x$

$9x + 6 = -12$

$-6 \quad -6$

$\frac{9x}{9} = \frac{-18}{9}$

$\boxed{x = -2}$

11. $6(m - 1) = 3(3m + 5)$

$6m - 6 = 9m + 15$

$-6m \quad -6m$

$-6 = 3m + 15$

$-15 \quad -15$

$\frac{-21}{3} = \frac{3m}{3}$

$\boxed{-7 = m}$

12. $7(p + 3) + 9 = 5(p - 2) - 3p$

$7p + 21 + 9 = 5p - 10 - 3p$

$7p + 30 = 2p - 10$

$-2p \quad -2p$

$5p + 30 = -10$

$-30 \quad -30$

$\frac{5p}{5} = \frac{-40}{5}$

$\boxed{p = -8}$

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NO SOLUTION & INFINITE SOLUTION

No Solution	Infinite Solution
$-4(2x+1) = -8x - 2$ $-8x - 4 = -8x - 2$ $+8x \quad +8x$ $-4 \neq -2$ $\emptyset$	$-5 - 9x = 3(1 - 3x) - 8$ $-5 - 9x = 3 - 9x - 8$ $-5 - 9x = -5 - 9x$ $+9x \quad +9x$ $-5 = -5$ ∞
There is **no possible number** that could work as a solution to the equation!	**Every number** could work as a solution!

MORE EXAMPLES!

1

$3(2x + 2) - 3x = 6 + 3x$

$6x + 6 - 3x = 6 + 3x$

$3x + 6 = 6 + 3x$

$-3x \quad -3x$

$6 = 6$

∞

2

$6(2x - 6) = -7(-2x + 4)$

$12x - 36 = 14x - 28$

$-12x \quad -12x$

$-36 = 2x - 28$

$+28 \quad +28$

$\frac{-8}{2} = \frac{2x}{2}$

$-4 = x$

3

$8(5x - 3) = 6(-3x - 4)$

$40x - 24 = -18x - 24$

$+18x \quad +18x$

$58x - 24 = -24$

$+24 \quad +24$

$\frac{58x}{58} = \frac{0}{58}$

$x = 0$

4

$3x - 13 = 7(x + 2) - 4(x - 7)$

$3x - 13 = 7x + 14 - 4x + 28$

$3x - 13 = 3x + 42$

$-3x \quad -3x$

$-13 \neq 42$

$\emptyset$

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Name: _____ **Unit 2:** Equations & Inequalities

Date: _____ **Bell:** _____ **Homework 3:** No Solution/Infinite Solution

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $2(n + 2) = 4n + 1 - 2n$

$2n + 4 = 2n + 1$

$-2n \quad -2n$

$4 \neq 1$

$\emptyset$

2. $-15 - 3w = 4w - 2w$

$-15 - 3w = 2w$

$+3w \quad +3w$

$\frac{-15}{5} = \frac{5w}{5}$

$-3 = w$

3. $2(x + 4) = 4x + 3 - 2x + 5$

$2x + 8 = 2x + 8$

$-2x \quad -2x$

$8 = 8$

∞

4. $4(2r - 1) = -2(3r + 16)$

$8r - 4 = -6r - 32$

$+6r \quad +6r$

$14r - 4 = -32$

$+4 \quad +4$

$14r = -28$

$r = -2$

5. $3a - 6a + 2 = 8a + 20 - 5a$

$-3a + 2 = 3a + 20$

$+3a \quad +3a$

$2 = 6a + 20$

$-20 \quad -20$

$\frac{-18}{6} = \frac{6a}{6}$

$a = -3$

6. $-(v + 4) + 5 = 4v + 1 - 5v$

$-v - 4 + 5 = -v + 1$

$-v + 1 = -v + 1$

$+v \quad +v$

$1 = 1$

∞

7. $8 - 6m = 4m + 48$

$+6m \quad +6m$

$8 = 10m + 48$

$-48 \quad -48$

$\frac{-40}{10} = \frac{10m}{10}$

$m = -4$

8. $11 + 3p - 7 = 6p + 5 - 3p$

$3p + 4 = 3p + 5$

$-3p \quad -3p$

$4 \neq 5$

$\emptyset$

9. $4y = 38 - \frac{1}{2}(4y + 16)$

$4y = 38 - 2y - 8$

$4y = -2y + 30$

$+2y \quad +2y$

$\frac{6y}{6} = \frac{30}{6}$

$y = 5$

10. $8x + 11 = 2(4x - 7) + 25$

$8x + 11 = 8x - 14 + 25$

$8x + 11 = 8x + 11$

$-8x \quad -8x$

$11 = 11$

∞

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
STEPS TO SOLVE	1. Cross multiply ($\frac{a}{b}=\frac{c}{d} \rightarrow a \cdot d = b \cdot c$) 2. Distribute (if necessary) 3. Solve!
GUIDED EXAMPLES	**1)** $\frac{4}{3}=\frac{x}{6}$ $3 \cdot x = 4 \cdot 6$; $\frac{3x}{3}=\frac{24}{3}$; $\boxed{x=8}$ **2)** $\frac{2}{10}=\frac{4}{n+5}$ $2(n+5)=10 \cdot 4$; $2n+10=40$; $-10 \quad -10$; $\frac{2n}{2}=\frac{30}{2}$; $\boxed{n=15}$ **3)** $\frac{y-3}{2}=\frac{y}{5}$ $2y=5(y-3)$; $2y=5y-15$; $-5y \quad -5y$; $\frac{-3y}{-3}=\frac{-15}{-3}$; $\boxed{y=5}$ **4)** $\frac{2k-3}{k+9}=\frac{3}{5}$ $5(2k-3)=3(k+9)$; $10k-15=3k+27$; $-3k \quad -3k$; $7k-15=27$; $+15 \quad +15$; $\frac{7k}{7}=\frac{42}{7}$; $\boxed{k=6}$
YOU TRY!	**Directions:** Solve each proportion. **1.** $\frac{x+10}{8}=\frac{x}{3}$ $8x=3(x+10)$; $8x=3x+30$; $-3x \quad -3x$; $\frac{5x}{5}=\frac{30}{5}$; $\boxed{x=6}$ **2.** $\frac{p+2}{p+7}=\frac{3}{6}$ $3(p+7)=6(p+2)$; $3p+21=6p+12$; $-3p \quad -3p$; $21=3p+12$; $-12 \quad -12$; $\frac{9}{3}=\frac{3p}{3}$; $\boxed{p=3}$ **3.** $\frac{w-2}{w}=\frac{3}{6}$ $3w=6(w-2)$; $3w=6w-12$; $-6w \quad -6w$; $\frac{-3w}{-3}=\frac{-12}{-3}$; $\boxed{w=4}$ **4.** $\frac{n-4}{6}=\frac{n-7}{9}$ $6(n-7)=9(n-4)$; $6n-42=9n-36$; $-6n \quad -6n$; $-42=3n-36$; $+36 \quad +36$; $\frac{-6}{3}=\frac{3n}{3}$; $\boxed{n=-2}$ **5.** $\frac{3}{2}=\frac{a-3}{a+1}$ $3(a+1)=2(a-3)$; $3a+3=2a-6$; $-2a \quad -2a$; $a+3=-6$; $-3 \quad -3$; $\boxed{a=-9}$ **6.** $\frac{25}{f+2}=\frac{5}{3}$ $5(f+2)=75$; $5f+10=75$; $-10 \quad -10$; $\frac{5f}{5}=\frac{65}{5}$; $\boxed{f=13}$

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7. $\frac{8}{m-9}=\frac{6}{m-8}$

$6(m-9)=8(m-8)$

$6m-54=8m-64$

$-6m \quad -6m$

$-54=2m-64$

$+64 \quad +64$

$\frac{10}{2}=\frac{2m}{2}$

$m=5$

8. $\frac{2}{r+4}=\frac{4}{r}$

$2r=4(r+4)$

$2r=4r+16$

$-4r \quad -4r$

$\frac{-2r}{-2}=\frac{16}{-2}$

$r=-8$

9. $\frac{5}{2}=\frac{v-8}{v-2}$

$5(v-2)=2(v-8)$

$5v-10=2v-16$

$-2v \quad -2v$

$3v-10=-16$

$+10 \quad +10$

$\frac{3v}{3}=\frac{-6}{3}$

$v=-2$

10. $\frac{6}{k-1}=\frac{8}{k-6}$

$6(k-6)=8(k-1)$

$6k-36=8k-8$

$-6k \quad -6k$

$-36=2k-8$

$+8 \quad +8$

$\frac{-28}{2}=\frac{2k}{2}$

$k=-14$

11. $\frac{y-7}{y-9}=\frac{3}{4}$

$4(y-7)=3(y-9)$

$4y-28=3y-27$

$-3y \quad -3y$

$y-28=-27$

$+28 \quad +28$

$y=1$

12. $\frac{h-6}{9}=\frac{h+9}{4}$

$4(h-6)=9(h+9)$

$4h-24=9h+81$

$-4h \quad -4h$

$-24=5h+81$

$-81 \quad -81$

$\frac{-105}{5}=\frac{5h}{5}$

$h=-21$

13. $\frac{r}{2}=\frac{r+10}{4}$

$4r=2(r+10)$

$4r=2r+20$

$-2r \quad -2r$

$\frac{2r}{2}=\frac{20}{2}$

$r=10$

14. $\frac{7}{2x-8}=\frac{3}{x-6}$

$7(x-6)=3(2x-8)$

$7x-42=6x-24$

$-6x \quad -6x$

$x-42=-24$

$+42 \quad +42$

$x=18$

15. $\frac{2}{3u+7}=\frac{3}{4u+9}$

$2(4u+9)=3(3u+7)$

$8u+18=9u+21$

$-8u \quad -8u$

$18=u+21$

$-21 \quad -21$

$-3=u$

16. $\frac{2x-1}{5}=\frac{3x+1}{10}$

$10(2x-1)=5(3x+1)$

$20x-10=15x+5$

$-15x \quad -15x$

$5x-10=5$

$+10 \quad +10$

$\frac{5x}{5}=\frac{15}{5}$

$x=3$

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Name: _____ **Unit 2:** Equations & Inequalities

Date: _____ **Bell:** _____ **Homework 4:** Proportions

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $\frac{x}{4}=\frac{3}{2}$

$2x = 4\cdot 3$

$\frac{2x}{2} = \frac{12}{2}$

$\boxed{x=6}$

2. $\frac{f+3}{12}=\frac{7}{2}$

$2(f+3) = 12\cdot 7$

$2f+6 = 84$

$-6 \quad -6$

$\frac{2f}{2} = \frac{78}{2}$

$\boxed{f=39}$

3. $\frac{4}{n+9}=\frac{2}{7}$

$2(n+9) = 4\cdot 7$

$2n+18 = 28$

$-18 \quad -18$

$\frac{2n}{2} = \frac{10}{2}$

$\boxed{n=5}$

4. $\frac{3}{8}=\frac{s-2}{24}$

$8(s-2) = 3\cdot 24$

$8s-16 = 72$

$+16 \quad +16$

$\frac{8s}{8} = \frac{88}{8}$

$\boxed{s=11}$

5. $\frac{3}{10}=\frac{b+7}{20}$

$10(b+7) = 3\cdot 20$

$10b+70 = 60$

$-70 \quad -70$

$\frac{10b}{10} = \frac{-10}{10}$

$\boxed{b=-1}$

6. $\frac{3}{16y}=\frac{1}{y+13}$

$16y = 3(y+13)$

$16y = 3y+39$

$-3y \quad -3y$

$\frac{13y}{13} = \frac{39}{13}$

$\boxed{y=3}$

7. $\frac{m}{3}=\frac{m+4}{7}$

$7m = 3(m+4)$

$7m = 3m+12$

$-3m \quad -3m$

$\frac{4m}{4} = \frac{12}{4}$

$\boxed{m=3}$

8. $\frac{a}{2}=\frac{a-4}{6}$

$2(a-4) = 6a$

$2a-8 = 6a$

$-2a \quad -2a$

$\frac{-8}{4} = \frac{4a}{4}$

$\boxed{a=-2}$

9. $\frac{x+2}{3}=\frac{x+4}{5}$

$5(x+2) = 3(x+4)$

$5x+10 = 3x+12$

$-3x \quad -3x$

$2x+10 = 12$

$-10 \quad -10$

$\frac{2x}{2} = \frac{2}{2}$

$\boxed{x=1}$

10. $\frac{n+8}{5}=\frac{n-1}{2}$

$2(n+8) = 5(n-1)$

$2n+16 = 5n-5$

$-2n \quad -2n$

$16 = 3n-5$

$+5 \quad +5$

$\frac{21}{3} = \frac{30}{3}$

$\boxed{n=7}$

11. $\frac{p-5}{p+1}=\frac{2}{3}$

$3(p-5) = 2(p+1)$

$3p-15 = 2p+2$

$-2p \quad -2p$

$p-15 = 2$

$+15 \quad +15$

$\boxed{p=17}$

12. $\frac{x+2}{6}=\frac{2x-10}{8}$

$8(x+2) = 6(2x-10)$

$8x+16 = 12x-60$

$-8x \quad -8x$

$16 = 4x-60$

$+60 \quad +60$

$76 = 4x$

$\boxed{x=19}$

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Properties of Equality

The steps to solve an equation are justified by properties of equality. They ensure that the equations stays balanced in order to solve for the missing variable.

- Addition Property of Equality
- Subtraction Property of Equality
- Multiplication Property of Equality
- Division Property of Equality
- Distributive Property

Directions: Identify which property justifies each step used to solve the equations below.

1

Equation Steps	Properties/Reasons
1. $2(x-9) = -10$	1. Given
2. $2x - 18 = -10$	2. Distributive Prop.
3. $2x = 8$	3. Addition Prop. of Eq.
4. $x = 4$	4. Division Prop. of Eq.

2

Equation Steps	Properties/Reasons
1. $\frac{m}{-3} + 10 = -1$	1. Given
2. $\frac{m}{-3} = -11$	2. Subtraction Prop. of Eq.
3. $m = 33$	3. Multiplication Prop. of Eq.

3

Equation Steps	Properties/Reasons
1. $\frac{5y-1}{2} = 7$	1. Given
2. $5y - 1 = 14$	2. Multiplication Prop. of Eq.
3. $5y = 15$	3. Addition Prop. of Eq.
4. $y = 3$	4. Division Prop. of Eq.

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4

Equation Steps	Properties/Reasons
1. $10k - 4 = 2k - 20$	1. Given
2. $8k - 4 = -20$	2. Subtraction Prop. of Eq.
3. $8k = -16$	3. Addition Prop. of Eq.
4. $k = -2$	4. Division Prop. of Eq.

5

Equation Steps	Properties/Reasons
1. $5n - 42 = 12n$	1. Given
2. $-42 = 7n$	2. Subtraction Prop. of Eq.
3. $-6 = n$	3. Division Prop. of Eq.
4. $n = -6$	4. Symmetric Prop.

6

Equation Steps	Properties/Reasons
1. $-8(w + 1) = -5(w + 10)$	1. Given
2. $-8w - 8 = -5w - 50$	2. Distributive Prop.
3. $-3w - 8 = -50$	3. Addition Prop. of Eq.
4. $-3w = -42$	4. Addition Prop. of Eq.
5. $w = 14$	5. Division Prop. of Eq.

7

Equation Steps	Properties/Reasons
1. $14 - 2(x + 8) = 5x - (3x - 34)$	1. Given
2. $14 - 2x - 16 = 5x - 3x + 34$	2. Distributive Prop.
3. $-2x - 2 = 2x + 34$	3. Combine Like Terms
4. $-4x - 2 = 34$	4. Subtraction Prop. of Eq.
5. $-4x = 36$	5. Addition Prop. of Eq.
6. $x = -9$	6. Division Prop. of Eq.

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Name: _____ Algebra I

Date: _____ Bell: _____ Unit 2: Equations & Inequalities

Quiz 2-1: Equations

Directions: Solve each equation. ***SHOW ALL WORK!***

1. $6y - 8 - 6 - 8y = 4$

$-2y - 14 = 4$

$+14 \quad +14$

$-2y = 18$

$\frac{-2y}{-2} = \frac{18}{-2}$

2. $-7(x + 7) = -56$

$-7x - 49 = -56$

$+49 \quad +49$

$\frac{-7x}{-7} = \frac{-7}{-7}$

3. $4(4m - 3) - (m - 5) = -52$

$16m - 12 - m + 5 = -52$

$15m - 7 = -52$

$+7 \quad +7$

$\frac{15m}{15} = \frac{-45}{15}$

4. $7 - 6v = -5 - 8v$

$+8v \quad +8v$

$7 + 2v = -5$

$-7 \quad -7$

$\frac{2v}{2} = \frac{-12}{2}$

5. $5(x + 6) - 2x = 3(x + 10)$

$5x + 30 - 2x = 3x + 30$

$3x + 30 = 3x + 30$

$-3x \quad -3x$

$30 = 30$

6. $9(k - 4) - 7k = 32 - 2(k + 8)$

$9k - 36 - 7k = 32 - 2k - 16$

$2k - 36 = -2k + 16$

$+2k \quad +2k$

$4k - 36 = 16$

$+36 \quad +36$

$\frac{4k}{4} = \frac{52}{4}$

7. $-6(w - 4) + 8w = 2(w + 9)$

$-6w + 24 + 8w = 2w + 18$

$2w + 24 = 2w + 18$

$-2w \quad -2w$

$24 \neq 18$

8. $-4(x + 5) = \frac{5}{3}(3x - 12)$

$-4x - 20 = 5x - 20$

$+4x \quad +4x$

$-20 = 9x - 20$

$+20 \quad +20$

$\frac{0}{9} = \frac{9x}{9}$

ANSWERS	
1.	$y = -9$
2.	$x = 1$
3.	$m = -3$
4.	$v = -6$
5.	∞
6.	$k = 13$
7.	$\emptyset$
8.	$x = 0$

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9. $\frac{4}{3}=\frac{8}{x-1}$

$4(x-1)=24$

$4x-4=24$

$+4 \quad +4$

$\frac{4x}{4}=\frac{28}{4}$

10. $\frac{5}{8}=\frac{n}{n-9}$

$8n=5(n-9)$

$8n=5n-45$

$-5n \quad -5n$

$\frac{3n}{3}=\frac{-45}{3}$

11. $\frac{5}{y-7}=\frac{3}{y-5}$

$5(y-5)=3(y-7)$

$5y-25=3y-21$

$-3y \quad -3y$

$2y-25=-21$

$+25 \quad +25$

$\frac{2y}{2}=\frac{4}{2}$

12. $\frac{3}{m-1}=\frac{6}{3m+8}$

$3(3m+8)=6(m-1)$

$9m+24=6m-6$

$-6m \quad -6m$

$3m+24=-6$

$-24 \quad -24$

$\frac{3m}{3}=\frac{-30}{3}$

Answers

9. $x=7$
10. $n=-16$
11. $y=2$
12. $m=-10$

B. $\frac{1}{4}\left(5x+\frac{16}{3}\right)-\frac{2}{3}\left(\frac{3}{4}x+\frac{1}{2}\right)=-5$



$\frac{5}{4}x+\frac{4}{3}-\frac{1}{2}x-\frac{1}{3}=-5$

$\frac{5}{4}x-\frac{2}{4}x+\frac{4}{3}-\frac{1}{3}=-5$

$\frac{3}{4}x+1=-5$

$-1 \quad -1$

$\frac{4}{3}\cdot\frac{3}{4}x=-6\cdot\frac{4}{3}$

$x=-8$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples

Steps to Solve

1. Make sure the absolute value expression is isolated.
2. Set the "inside" equal to both the positive and negative number from the other side of the equation.
3. Solve both equations.
4. Write your answer as a solution set. $x = \{_, _\}$

Examples

Directions: Solve each equation. Write your answer as a solution set.

1. $|x| = 2$

$x = 2 \quad x = -2$

$x = \{-2, 2\}$

2. $|m| = 14$

$m = 14 \quad m = -14$

$m = \{-14, 14\}$

3. $|5z| = 40$

$\frac{5z}{5} = \frac{40}{5} \quad \frac{5z}{5} = \frac{-40}{5}$

$z = 8 \quad z = -8$

$z = \{-8, 8\}$

4. $|-7a| = 28$

$\frac{-7a}{-7} = \frac{28}{-7} \quad \frac{-7a}{-7} = \frac{-28}{-7}$

$a = -4 \quad a = 4$

$a = \{-4, 4\}$

5. $|d + 1| = 8$

$d + 1 = 8 \quad d + 1 = -8$

$-1 \quad -1 \quad -1 \quad -1$

$d = 7 \quad d = -9$

$d = \{-9, 7\}$

6. $|w - 3| = 4$

$w - 3 = 4 \quad w - 3 = -4$

$+3 \quad +3 \quad +3 \quad +3$

$w = 7 \quad w = -1$

$w = \{-1, 7\}$

7. $|4n + 2| = 34$

$4n + 2 = 34 \quad 4n + 2 = -34$

$-2 \quad -2 \quad -2 \quad -2$

$\frac{4n}{4} = \frac{32}{4} \quad \frac{4n}{4} = \frac{-36}{4}$

$n = 8 \quad n = -9$

$n = \{-9, 8\}$

8. $|-2v - 5| = 17$

$-2v - 5 = 17 \quad -2v - 5 = -17$

$+6 \quad +5 \quad +5 \quad +5$

$\frac{-2v}{-2} = \frac{22}{-2} \quad \frac{-2v}{-2} = \frac{-12}{-2}$

$v = -11 \quad v = 6$

$v = \{-11, 6\}$

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What if there is "stuff" outside the Absolute Value Bars?

Isolate the abs. value!

9. $|c| + 2 = 12$

$-2 \quad -2$

$|c| = 10$

$c = 10 \qquad c = -10$

$\boxed{c = \{-10, 10\}}$

10. $|y| - 8 = -5$

$+8 \quad +8$

$|y| = 3$

$y = 3 \qquad y = -3$

$\boxed{y = \{-3, 3\}}$

11. $\frac{6|x|}{6} = \frac{24}{6}$

$|x| = 4$

$x = 4 \qquad x = -4$

$\boxed{X = \{-4, 4\}}$

12. $\frac{-2|z+3|}{-2} = \frac{-14}{-2}$

$|z+3| = 7$

$z+3 = 7 \qquad z+3 = -7$

$-3 \quad -3 \qquad -3 \quad -3$

$z = 4 \qquad z = -10$

$\boxed{z = \{-10, 4\}}$

13. $6\frac{|2w-4|}{6} = 3 \cdot 6$

$|2w-4| = 18$

$2w-4 = 18 \qquad 2w-4 = -18$

$+4 \quad +4 \qquad +4 \quad +4$

$\frac{2w}{2} = \frac{22}{2} \qquad \frac{2w}{2} = \frac{-14}{2}$

$w = 11 \qquad w = -7$

$\boxed{w = \{-7, 11\}}$

14. $-10|h+5| - 3 = -83$

$+3 \quad +3$

$\frac{-10|h+5|}{-10} = \frac{-80}{-10}$

$|h+5| = 8$

$h+5 = 8 \qquad h+5 = -8$

$-5 \quad -5 \qquad -6 \quad -5$

$h = 3 \qquad h = -13$

$\boxed{h = \{-13, 3\}}$

15. $7|4n+8| - 6 = 106$

$+6 \quad +6$

$\frac{7|4n+8|}{7} = \frac{112}{7}$

$|4n+8| = 16$

$4n+8 = 16 \qquad 4n+8 = -16$

$4n = 8 \qquad 4n = -24$

$n = 2 \qquad n = -6$

$\boxed{n = \{-6, 2\}}$

16. $\frac{3}{2}|4r-4| - 24 = -21$

$+24 \quad +24$

$\frac{3}{2}|4r-4| = 3$

$\frac{2}{3} \cdot \frac{3}{2}|4r-4| = 3 \cdot \frac{2}{3}$

$|4r-4| = 2$

$4r-4 = 2 \qquad 4r-4 = -2$

$4r = 6 \qquad 4r = 2$

$r = 3/2 \qquad r = 1/2$

$\boxed{r = \{1/2, 3/2\}}$

Special Cases

Directions: Solve the equations below. Explain their solutions.

17. $|x| = -6$

$\emptyset$

18. $\frac{-3|2x+1|}{-3} = \frac{21}{-3}$

$|2x+1| = -7$

Explain:

The absolute value cannot equal a negative number once isolated.

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Name: _____ **Unit 2:** Equations & Inequalities

Date: _____ Bell: _____ **Homework 5:** Absolute Value Equations

**** This is a 2-page document! ****

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $|x| = 3$

$x = 3$ $\quad x = -3$

$x = \{-3, 3\}$

2. $|m| = -10$

$\emptyset$

3. $|-6k| = 30$

$\frac{-6k}{-6} = \frac{30}{-6}$ $\quad \frac{-6k}{-6} = \frac{-30}{-6}$

$k = -5$ $\quad k = 5$

$k = \{-5, 5\}$

4. $|c - 10| = 3$

$c - 10 = 3$ $\quad c - 10 = -3$

$+10 \ +10$ $\quad +10 \ +10$

$c = 13$ $\quad c = 7$

$c = \{7, 13\}$

5. $|2k + 3| = 7$

$2k + 3 = 7$ $\quad 2k + 3 = -7$

$-3 \ -3$ $\quad -3 \ -3$

$\frac{2k}{2} = \frac{4}{2}$ $\quad \frac{2k}{2} = \frac{-10}{2}$

$k = 2$ $\quad k = -5$

$k = \{-5, 2\}$

6. $|-4g - 2| = 10$

$-4g - 2 = 10$ $\quad -4g - 2 = -10$

$+2 \ +2$ $\quad +2 \ +2$

$\frac{-4g}{-4} = \frac{12}{-4}$ $\quad \frac{-4g}{-4} = \frac{-8}{-4}$

$g = -3$ $\quad g = 2$

$g = \{-3, 2\}$

7. $\left|\frac{1}{2}x + 5\right| = 13$

$\frac{1}{2}x + 5 = 13$ $\quad \frac{1}{2}x + 5 = -13$

$-5 \ -5$ $\quad -5 \ -5$

$2 \cdot \frac{1}{2}x = 8 \cdot 2$ $\quad 2 \cdot \frac{1}{2}x = -18 \cdot 2$

$x = 16$ $\quad x = -36$

$x = \{-36, 16\}$

8. $|26 - 4c| = -2$

$\emptyset$

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9. $\frac{-3|p|}{-3} = \frac{12}{-3}$

$|p| = -4$

$\emptyset$

10. $|m| + 2 = 11$

$-2 \quad -2$

$|m| = 9$

$m = 9 \qquad m = -9$

$m = \{-9, 9\}$

11. $\frac{|4x-12|}{2} = 8 \cdot 2$

$|4x-12| = 16$

$4x - 12 = 16$ $\qquad$ $4x - 12 = -16$

$+12 \quad +12$ $\qquad$ $+12 \quad +12$

$\frac{4x}{4} = \frac{28}{4}$ $\qquad$ $\frac{4x}{4} = \frac{-4}{4}$

$x = 7$ $\qquad$ $x = -1$

$x = \{-1, 7\}$

12. $|3b+9| - 16 = -52$

$+16 \quad +16$

$|3b+9| = -36$

$\emptyset$

13. $\frac{3}{4} \cdot \frac{4}{3}|3w-6| = 16 \cdot \frac{3}{4}$

$|3w-6| = 12$

$3w - 6 = 12$ $\qquad$ $3w - 6 = -12$

$+6 \quad +6$ $\qquad$ $+6 \quad +6$

$\frac{3w}{3} = \frac{18}{3}$ $\qquad$ $\frac{3w}{3} = \frac{-6}{3}$

$w = 6$ $\qquad$ $w = -2$

$w = \{-2, 6\}$

14. $\frac{-5|6a+21|}{-5} = \frac{-15}{-5}$

$|6a+21| = 3$

$6a + 21 = 3$ $\qquad$ $6a + 21 = -3$

$-21 \quad -21$ $\qquad$ $-21 \quad -21$

$\frac{6a}{6} = \frac{-18}{6}$ $\qquad$ $\frac{6a}{6} = \frac{-24}{6}$

$a = -3$ $\qquad$ $a = -4$

$a = \{-3, -4\}$

15. $4|v-2| - 3 = 25$

$+3 \quad +3$

$\frac{4|v-2|}{4} = \frac{28}{4}$

$|v-2| = 7$

$v - 2 = 7$ $\qquad$ $v - 2 = -7$

$+2 \quad +2$ $\qquad$ $+2 \quad +2$

$v = 9$ $\qquad$ $v = -5$

$v = \{-5, 9\}$

16. $3 - \frac{1}{2}|r+9| = -11$

$-3 \qquad -3$

$-2 \cdot -\frac{1}{2}|r+9| = -14 \cdot -2$

$|r+9| = 28$

$r + 9 = 28$ $\qquad$ $r + 9 = -28$

$-9 \quad -9$ $\qquad$ $-9 \quad -9$

$r = 19$ $\qquad$ $r = -37$

$r = \{-37, 19\}$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples

Literal Equations

Equations with multiple variables.

SOLVE EACH OF THE EQUATIONS BELOW FOR x:

$2x - 5 = 13$
$+5 \quad +5$
$\frac{2x}{2} = \frac{18}{2}$
$\boxed{x = 9}$

$ax - b = c$
$+b \quad +b$
$\frac{ax}{a} = \frac{c+b}{a}$
$\boxed{x = \frac{c+b}{a}}$

Quick Write



Identify the similarities and differences between the two equations:

- Both use opposite operations to isolate
- The literal equation doesn't simplify nicely, due to multiple variables.

One-Step Problems

1. $A = lw$ — solve for w
$\frac{A}{l} = \frac{lw}{l}$
$\boxed{w = \frac{A}{l}}$

2. $A = bh$ — solve for h
$\frac{A}{b} = \frac{bh}{b}$
$\boxed{h = \frac{A}{b}}$

3. $d = rt$ — solve for t
$\frac{d}{r} = \frac{rt}{r}$
$\boxed{t = \frac{d}{r}}$

4. $I = prt$ — solve for t
$\frac{I}{pr} = \frac{prt}{pr}$
$\boxed{t = \frac{I}{pr}}$

5. $V = lwh$ — solve for h
$\frac{V}{lw} = \frac{lwh}{lw}$
$\boxed{h = \frac{V}{lw}}$

6. $A = \pi r^2$ — solve for r
$\frac{A}{\pi} = \frac{\pi r^2}{\pi}$
$\sqrt{\frac{A}{\pi}} = \sqrt{r^2}$
$\boxed{r = \pm\sqrt{\frac{A}{\pi}}}$

7. $m = c - s$ — solve for c
$+s \quad +s$
$\boxed{c = m + s}$

8. $D = \frac{m}{v} \cdot V$ — solve for m
$\boxed{m = Dv}$

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Multi-Step Problems

Helpful Hints:

- Think backwards ← PEMDAS
- Remove fractions by multiplying by the reciprocal.
- Last step is USUALLY to divide by whatever is next to your variable.

9. $A = \frac{1}{2}bh$ solve for h · 2

$\frac{2A}{b} = \frac{bh}{b}$

$\boxed{\frac{2A}{b} = h}$

10. $V = \frac{1}{3}Bh$ solve for B · 3

$\frac{3V}{h} = \frac{Bh}{h}$

$\boxed{B = \frac{3V}{h}}$

11. $K = \frac{mv^2}{2}$ solve for m · 2

$\frac{2K}{v^2} = \frac{mv^2}{v^2}$

$\boxed{\frac{2K}{v^2} = m}$

12. $a = \frac{b+c}{d}$ solve for b · d

$ad = b + c$

$-c \quad -c$

$\boxed{ad - c = b}$

13. $P = 2L + 2W$ solve for W

$-2L \quad -2L$

$\frac{P-2L}{2} = \frac{2W}{2}$

$\boxed{\frac{P-2L}{2} = W}$

14. $Ax + By = C$ solve for y

$-Ax \quad -Ax$

$\frac{By}{B} = \frac{C-Ax}{B}$

$\boxed{y = \frac{C-Ax}{B}}$

15. $y = mx + b$ solve for x

$-b \quad -b$

$\frac{y-b}{m} = \frac{mx}{m}$

$\boxed{\frac{y-b}{m} = x}$

16. $A = P + Prt$ solve for t

$-P \quad -P$

$\frac{A-P}{Pr} = \frac{Prt}{Pr}$

$\boxed{\frac{A-P}{Pr} = t}$

17. $C = \frac{5}{9}(F-32)$ solve for F · $\frac{9}{5}$

$\frac{9}{5}C = F - 32$

$+32 \quad +32$

$\boxed{\frac{9}{5}C + 32 = F}$

18. $A = \frac{1}{2}h(b_1 + b_2)$ solve for b_1 · 2

$\frac{2A}{h} = \frac{h(b_1+b_2)}{h}$

$\frac{2A}{h} = b_1 + b_2$

$-b_2 \quad -b_2$

$\boxed{b_1 = \frac{2A}{h} - b_2}$

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Name: _____ **Unit 2:** Equations & Inequalities

Date: _____ **Bell:** _____ **Homework 6:** Literal Equations

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $\frac{d}{2} = \frac{2r}{2}$ solve for r

$\frac{d}{2} = r$

2. $s = \frac{2d}{t} \cdot t$ solve for d

$\frac{st}{2} = \frac{2d}{2}$

$\frac{st}{2} = d$

3. $\frac{C}{2\pi} = \frac{2\pi r}{2\pi}$ solve for r

$\frac{C}{2\pi} = r$

4. $\frac{V}{r^2h} = \frac{\pi r^2 h}{r^2 h}$ solve for π

$\frac{V}{r^2h} = \pi$

5. $\frac{I}{pt} = \frac{prt}{pt}$ solve for r

$\frac{I}{pt} = r$

6. $\frac{E}{m} = \frac{mc^2}{m}$ solve for c

$\sqrt{\frac{E}{m}} = \sqrt{c^2}$

$\pm\sqrt{\frac{E}{m}} = c$

7. $P = \frac{100a}{t} \cdot t$ solve for a

$\frac{Pt}{100} = \frac{100a}{100}$

$\frac{Pt}{100} = a$

8. $V = \frac{1}{3}\pi r^2 h \cdot 3$ solve for h

$\frac{3V}{\pi r^2} = \frac{\pi r^2 h}{\pi r^2}$

$\frac{3V}{\pi r^2} = h$

9. $y = mx + b$ solve for m

$-b \quad -b$

$\frac{y-b}{x} = \frac{mx}{x}$

$\frac{y-b}{x} = m$

10. $\frac{a}{2} = \frac{2(b-c)}{2}$ solve for b

$\frac{a}{2} = b - c$

$+c \qquad +c$

$\frac{a}{2} + c = b$

11. $w = 3x + 7y$ solve for y

$-3x \quad -3x$

$\frac{w-3x}{7} = \frac{7y}{7}$

$\frac{w-3x}{7} = y$

12. $A = \frac{1}{2}h(b_1 + b_2) \cdot 2$ solve for b_2

$\frac{2A}{h} = \frac{h(b_1+b_2)}{h}$

$\frac{2A}{h} = b_1 + b_2$

$-b_1 \quad -b_1$

$\frac{2A}{h} - b_1 = b_2$

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equations review!

Find Someone Who...

Name: _____

Date: _____ Block: _____

Choose one equation from each topic to solve on your own.
Then, find new people to solve the others. SHOW ALL WORK!



topic 1: multi-step equations

1. $6(x + 3) - (x + 4) = -11$

$6x + 18 - x - 4 = -11$

$5x + 14 = -11$

$-14 \quad -14$

$\frac{5x}{5} = \frac{-25}{5}$

$\boxed{x = -5}$

Solved by: _____

2. $-8(n - 4) = n - 40$

$-8n + 32 = n - 40$

$+8n \quad +8n$

$32 = 9n - 40$

$+40 \quad +40$

$\frac{72}{9} = \frac{9n}{9}$

$\boxed{n = 8}$

Solved by: _____

3. $7(y - 5) + 3y = 10(y + 1)$

$7y - 35 + 3y = 10y + 10$

$10y - 35 = 10y + 10$

$-10y \quad -10y$

$-35 \neq 10$

$\boxed{\emptyset}$

Solved by: _____

4. $-5(k + 2) = 13 - (k - 1)$

$-5k - 10 = 13 - k + 1$

$-5k - 10 = -k + 14$

$+5k \quad +5k$

$-10 = 4k + 14$

$-14 \quad -14$

$\frac{-24}{4} = \frac{4k}{4}$

$\boxed{k = -6}$

Solved by: _____

5. $-6 - (2x + 7) = -2(x + 6) - 1$

$-6 - 2x - 7 = -2x - 12 - 1$

$-2x - 13 = -2x - 13$

$+2x \quad +2x$

$-13 = -13$

$\boxed{\infty}$

Solved by: _____

6. $4(2x + 7) = 12 - 2(10 - 2x)$

$8x + 28 = 12 - 20 + 4x$

$8x + 28 = 4x - 8$

$-4x \quad -4x$

$4x + 28 = -8$

$-28 \quad -28$

$\frac{4x}{4} = \frac{-36}{4}$

$\boxed{x = -9}$

Solved by: _____

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topic 2: proportions

7. $\frac{5}{x-3}=\frac{7}{x-9}$

$5(x-9)=7(x-3)$

$5x-45=7x-21$

$-5x \quad -5x$

$-45=2x-21$

$+21 \quad +21$

$\frac{-24}{2}=\frac{2x}{2}$

$x=-12$

Solved by: _____

8. $\frac{5}{6}=\frac{x-10}{3x+6}$

$6(x-10)=5(3x+6)$

$6x-60=15x+30$

$-6x \quad -6x$

$-60=9x+30$

$-30 \quad -30$

$\frac{-90}{9}=\frac{9x}{9}$

$x=-10$

Solved by: _____

topic 3: absolute value equations

9. $|x|=9$

$x=9 \qquad x=-9$

$x=\{-9,9\}$

Solved by: _____

10. $|6c-3|=21$

$6c-3=21$ $\qquad$ $6c-3=-21$

$+3 \quad +3$ $\qquad$ $+3 \quad +3$

$\frac{6c}{6}=\frac{24}{6}$ $\qquad$ $\frac{6c}{6}=\frac{-18}{6}$

$c=4 \qquad c=-3$

$c=\{-3,4\}$

Solved by: _____

11. $|7m|+4=25$

$-4 \quad -4$

$|7m|=21$

$\frac{7m}{7}=\frac{21}{7}$ $\qquad$ $\frac{7m}{7}=\frac{-21}{7}$

$m=3 \qquad m=-3$

$m=\{-3,3\}$

Solved by: _____

12. $5|2n+8|+3=33$

$-3 \quad -3$

$\frac{5|2n+8|}{5}=\frac{30}{5}$

$|2n+8|=6$

$2n+8=6$ $\qquad$ $2n+8=-6$

$2n=-2 \qquad 2n=-14$

$n=-1 \qquad n=-7$

$n=\{-7,-1\}$

Solved by: _____

topic 4: multi-variable equations

13. Solve $w=\frac{1}{2}x^2y^2$ for y

$\frac{2w}{x^2}=\frac{x^2y}{x^2}$

$\frac{2w}{x^2}=y$

Solved by: _____

14. Solve $c=8a-3b$ for a

$+3b \quad +3b$

$\frac{c+3b}{8}=\frac{8a}{8}$

$\frac{c+3b}{8}=a$

Solved by: _____

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Name: _____ **Unit 2:** Equations & Inequalities

Date: _____ Bell: _____ **Homework 7:** Equations Review

**** This is a 2-page document! ****

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $5(n+4)-(n+7)=-11$

$5n+20-n-7=-11$

$4n+13=-11$

$-13 \quad -13$

$\frac{4n}{4}=\frac{-24}{4}$

$n=-6$

2. $2(x-8)+4x=6(x-2)-4$

$2x-16+4x=6x-12-4$

$6x-16=6x-16$

$-6x \quad -6x$

$-16=-16$

∞

3. $2(k+4)=3(k-2)$

$2k+8=3k-6$

$-2k \quad -2k$

$8=k-6$

$+6 \quad +6$

$14=k$

4. $17-3(w+5)=6(w+5)-2w$

$17-3w-15=6w+30-2w$

$-3w+2=4w+30$

$+3w \quad +3w$

$2=7w+30$

$-30 \quad -30$

$\frac{-28}{7}=\frac{7w}{7}$

$w=-4$

5. $9y-(2y-3)=5(y-2)+2y$

$9y-2y+3=5y-10+2y$

$7y+3=7y-10$

$-7y \quad -7y$

$3 \neq -10$

$\emptyset$

6. $-8(a-4)+3a=2(4a+9)+1$

$-8a+32+3a=8a+18+1$

$-5a+32=8a+19$

$+5a \quad +5a$

$32=13a+19$

$-19 \quad -19$

$\frac{13}{13}=\frac{13a}{13}$

$a=1$

7. $\frac{n-6}{3}=\frac{n-2}{9}$

$9(n-6)=3(n-2)$

$9n-54=3n-6$

$-3n \quad -3n$

$6n-54=-6$

$+54 \quad +54$

$\frac{6n}{6}=\frac{48}{6}$

$n=8$

8. $\frac{10}{2x+4}=\frac{4}{x+5}$

$10(x+5)=4(2x+4)$

$10x+50=8x+16$

$-8x \quad -8x$

$2x+50=16$

$-50 \quad -50$

$\frac{2x}{2}=\frac{-34}{2}$

$x=-17$

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9. $|x + 9| = 7$

$x+9=7$ $\quad -9\ -9$

$x=-2$

$x+9=-7$ $\quad -9\ -9$

$x=-16$

$\boxed{x=\{-16,-2\}}$

10. $|2m - 1| = 3$

$2m-1=3$ $\quad +1\ +1$

$\frac{2m}{2}=\frac{4}{2}$

$m=2$

$2m-1=-3$ $\quad +1\ +1$

$\frac{2m}{2}=\frac{-2}{2}$

$m=-1$

$\boxed{m=\{-1,2\}}$

11. $|3g| - 4 = 11$ $\quad +4\ +4$

$|3g|=15$

$\frac{3g}{3}=\frac{15}{3}$

$g=5$

$\frac{3g}{3}=\frac{-15}{3}$

$g=-5$

$\boxed{g=\{-5,5\}}$

12. $3|2v - 7| - 4 = -1$ $\quad +4\ +4$

$\frac{3|2v-7|}{3}=\frac{3}{3}$

$|2v-7|=1$

$2v-7=1$

$2v=8$

$v=4$

$2v-7=-1$

$2v=6$

$v=3$

$\boxed{v=\{3,4\}}$

13. Solve $A=\frac{1}{2}bh$ for b $\quad \cdot 2$

$\frac{2A}{h}=\frac{bh}{h}$

$\boxed{\frac{2A}{h}=b}$

14. Solve $P = 2l + 2w$ for l

$-2w\ \ -2w$

$\frac{P-2w}{2}=\frac{2l}{2}$

$\boxed{\frac{P-2w}{2}=l}$

15. Solve $s = n(a + 1)$ for a

$\frac{s}{n}=a+1$ $\quad -1$

$\boxed{\frac{s}{n}-1=a}$

16. Solve $\frac{1}{3}(2x - y) = z$ for x $\quad \cdot 3$

$2x-y=3z$ $\quad +y\ \ +y$

$\frac{2x}{2}=\frac{3z+y}{2}$

$\boxed{x=\frac{3z+y}{2}}$

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Using Algebra to Solve Word Problems

1. **DEFINE A VARIABLE** — Use "LET STATEMENTS" to define your variable.
2. **SET UP EQUATION & SOLVE** — Translate into an equation using using your let statements. Then solve!
3. **DEFINE ANSWER** — Give exactly what the problem is asking for.

Set I: Finding Two Numbers

1. The larger of two numbers is four more than the smaller number. If the sum of the numbers is 74, find the numbers.

let x = smaller #
let $x+4$ = larger #

$x + x + 4 = 74$
$2x + 4 = 74$
$-4 \quad -4$
$\frac{2x}{2} = \frac{70}{2}$
$x = 35$

35, 39

2. The larger of two numbers is seven less than three times the smaller number. If the sum of the numbers is 61, find the numbers.

let x = smaller #
let $3x-7$ = larger #

$x + 3x - 7 = 61$
$4x - 7 = 61$
$+7 \quad +7$
$\frac{4x}{4} = \frac{68}{4}$
$x = 17$

17, 44

3. The larger of two numbers is one more than four times the smaller number. If the sum of the numbers is 106, find the numbers.

let x = smaller #
let $4x+1$ = larger #

$x + 4x + 1 = 106$
$5x + 1 = 106$
$-1 \quad -1$
$\frac{5x}{5} = \frac{105}{5}$
$x = 21$

21, 85

4. The larger of two numbers is twice the smaller number. If 12 is subtracted from the larger number, the result is 7 more than the smaller number. Find the numbers.

let x = smaller #
let $2x$ = larger #

$2x - 12 = x + 7$
$-x \quad -x$
$x - 12 = 7$
$+12 \quad +12$
$x = 19$

19, 38

5. The highest score on an Algebra test was 42 points more than the lowest. When added together, the lowest and highest score was 154. Find both the highest and lowest score.

let x = lowest score
let $x+42$ = highest score

$x + x + 42 = 154$
$2x + 42 = 154$
$-42 \quad -42$
$\frac{2x}{2} = \frac{112}{2}$
$x = 56$

56, 98

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6. Garrett has one dollar less than three times as much money as Liz has. Together they have $179. How much money does Garrett have?

let x = Liz's $

let $3x-1$ = Garrett's $

$x + 3x - 1 = 179$

$4x - 1 = 179$

$+1 \quad +1$

$\frac{4x}{4} = \frac{180}{4}$

$x = 45$

$134

7. The Buffalo Bills scored 24 more than twice the number of points that the Miami Dolphins scored. Altogether, the teams scored 66 points. How many did each team score individually?

let x = Dolphins' score

let $2x+24$ = Bills' score

$x + 2x + 24 = 66$

$3x + 24 = 66$

$-24 \quad -24$

$\frac{3x}{3} = \frac{42}{3}$

$x = 14$

Dolphins: 14 pts
Bills: 52 pts

8. There are 10 less red skittles than orange skittles in the bag. The orange skittles are also twice the number of red skittles. Find the number of red and orange skittles in the bag.

let x = orange

let $x-10$ = red

$x = 2(x-10)$

$x = 2x - 20$

$-2x \quad -2x$

$\frac{-x}{-1} = \frac{-20}{-1}$

$x = 20$

20 orange and 10 red Skittles

Set 2: Perimeter of a Rectangle ($P = 2l + 2w$)

9. The length of a rectangle is six inches more than its width. If the perimeter of the rectangle is 24 inches, find its dimensions.

let x = width

let $x+6$ = length

$2(x) + 2(x+6) = 24$

$2x + 2x + 12 = 24$

$4x + 12 = 24$

$-12 \quad -12$

$\frac{4x}{4} = \frac{12}{4}$

$x = 3$

3in, 9in

10. The length of a rectangle is five inches more than four times its width. If the perimeter of the rectangle is 90 inches, find its dimensions.

let x = width

let $4x+5$ = length

$2(x) + 2(4x+5) = 90$

$2x + 8x + 10 = 90$

$10x + 10 = 90$

$-10 \quad -10$

$\frac{10x}{10} = \frac{80}{10}$

$x = 8$

8in, 37in

11. The length of a rectangle is three centimeters less than twice its width. If the perimeter of the rectangle is 18 centimeters, find its dimensions.

let x = width

let $2x-3$ = length

$2(x) + 2(2x-3) = 18$

$2x + 4x - 6 = 18$

$6x - 6 = 18$

$+6 \quad +6$

$\frac{6x}{6} = \frac{24}{6}$

$x = 4$

4cm, 5cm

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Name: _____ **Unit 2:** Equations & Inequalities

Date: _____ **Bell:** _____ **Homework 8:** Word Problems

**** This is a 2-page document! ****

Directions: Solve each problem using the 3-step process.

1. The larger of two numbers is 12 more than the smaller number. If the sum of the two numbers is 74, find the two numbers.

let x = smaller #
let $x+12$ = larger #

$x + x + 12 = 74$
$2x + 12 = 74$
$-12 \quad -12$
$\frac{2x}{2} = \frac{62}{2}$ $x = 31$

31, 43

2. The larger of two numbers is 10 less than twice the smaller number. If the sum of the two numbers is 38, find the two numbers.

let x = smaller #
let $2x-10$ = larger #

$x + 2x - 10 = 38$
$3x - 10 = 38$
$+10 \quad +10$
$\frac{3x}{3} = \frac{48}{3}$ $x = 16$

16, 22

3. The number of years that Brad has been on the soccer team is 2 less than 5 times the number of years that Scott has. In total, the boys have been on the soccer team for 10 years. How long has Brad been on the soccer team?

let x = Scott's years
let $5x-2$ = Brad's years

$x + 5x - 2 = 10$
$6x - 2 = 10$
$+2 \quad +2$
$\frac{6x}{6} = \frac{12}{6}$ $x = 2$

8 years

4. There are 10 less trumpet players than saxophone players. The number of saxophone players is 3 times the number of trumpet players. How many trumpet players are there?

let x = saxophone players
let $x-10$ = trumpet players

$x = 3(x-10)$
$x = 3x - 30$
$-3x \quad -3x$
$\frac{-2x}{-2} = \frac{-30}{-2}$ $x = 15$

5 trumpet players

5. The length of a rectangle is 3 inches more than its width. If the perimeter is 42 inches, find the dimensions of the rectangle.

let x = width
let $x+3$ = length

$2(x) + 2(x+3) = 42$
$2x + 2x + 6 = 42$
$4x + 6 = 42$
$-6 \quad -6$
$\frac{4x}{4} = \frac{36}{4}$ $x = 9$

9 in, 12 in

6. The length of a rectangle is 4 centimeters more than twice its width. If the perimeter of the rectangle is 86 centimeters, find the dimensions of the rectangle.

let x = width
let $2x+4$ = length

$2(x) + 2(2x+4) = 86$
$2x + 4x + 8 = 86$
$6x + 8 = 86$
$-8 \quad -8$
$\frac{6x}{6} = \frac{78}{6}$ $x = 13$

13 cm, 30 cm

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7. The length of a rectangle is 5 less than twice its width. The perimeter is 26 meters. Find the dimensions of the rectangle.

let x = width
let $2x-5$ = length

$2(x) + 2(2x-5) = 26$
$2x + 4x - 10 = 26$
$6x - 10 = 26$
$+10 \quad +10$
$\frac{6x}{6} = \frac{36}{6}$
$x = 6$

6m, 7m

8. Find two consecutive numbers whose sum is 115.

let x = 1st consec #
let $x+1$ = 2nd consec #

$x + x + 1 = 115$
$2x + 1 = 115$
$-1 \quad -1$
$\frac{2x}{2} = \frac{114}{2}$
$x = 57$

57, 58

9. Find two consecutive **odd** numbers whose sum is 144.

let x = 1st consec odd #
let $x+2$ = 2nd consec odd #

$x + x + 2 = 144$
$2x + 2 = 144$
$-2 \quad -2$
$\frac{2x}{2} = \frac{142}{2}$
$x = 71$

71, 73

10. Find two consecutive **even** numbers whose sum is 126.

let x = 1st consec even #
let $x+2$ = 2nd consec even #

$x + x + 2 = 126$
$2x + 2 = 126$
$-2 \quad -2$
$\frac{2x}{2} = \frac{124}{2}$
$x = 62$

62, 64

11. Find three consecutive numbers whose sum is 84.

let x = 1st consec #
let $x+1$ = 2nd consec #
let $x+2$ = 3rd consec #

$x + x + 1 + x + 2 = 84$
$3x + 3 = 84$
$-3 \quad -3$
$\frac{3x}{3} = \frac{81}{3}$
$x = 27$

27, 28, 29

12. Find three consecutive **even** integers such that the sum of the smallest number and twice the middle number is 20 more than the largest number.

let x = 1st consec even #
let $x+2$ = 2nd consec even #
let $x+4$ = 3rd consec even #

$x + 2(x+2) = x + 4 + 20$
$x + 2x + 4 = x + 24$
$3x + 4 = x + 24$
$-x \quad -x$
$2x + 4 = 24$
$-4 \quad -4$
$\frac{2x}{2} = \frac{20}{2}$
$x = 10$

10, 12, 14

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Group Members: Per: _____



REVIEW: Equations & Word Problems

Directions: Work together to solve each problem. **SHOW ALL WORK!** Each person should be participating. At the end of class, one person's paper will be chosen at random and graded for the group.

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $-2(4x+3)-(x-7)=46$

$-8x-6-x+7=46$

$-9x+1=46$

$-1 \quad -1$

$\frac{-9x}{-9}=\frac{45}{-9}$

$x=-5$

2. $-\frac{4}{3}(12k+27)=-57-9k$

$-16k-36=-57-9k$

$+16k \qquad +16k$

$-36=-57+7k$

$+57 \quad +57$

$\frac{21}{7}=\frac{7k}{7}$

$k=3$

3. $23-2(5a+9)=5(2a-11)$

$23-10a-18=10a-55$

$-10a+5=10a-55$

$+10a \qquad +10a$

$5=20a-55$

$+55 \quad +55$

$\frac{60}{20}=\frac{20a}{20}$

$a=3$

4. $\frac{5}{2}(6w-16)=18w-(3w+40)$

$15w-40=18w-3w-40$

$15w-40=15w-40$

$-15w \qquad -15w$

$-40=-40$

∞

5. $\frac{10}{a-6}=\frac{5}{6}$

$6(10)=5(a-6)$

$60=5a-30$

$+30 \qquad +30$

$\frac{90}{5}=\frac{5a}{5}$

$a=18$

6. $\frac{g}{8}=\frac{g-1}{10}$

$10g=8(g-1)$

$10g=8g-8$

$-8g \quad -8g$

$\frac{2g}{2}=\frac{-8}{2}$

$g=-4$

7. $\frac{z-9}{6}=\frac{z+7}{3}$

$3(z-9)=6(z+7)$

$3z-27=6z+42$

$-3z \qquad -3z$

$-27=3z+42$

$-42 \qquad -42$

$-69=3z$

$z=-23$

8. $\frac{5}{2}=\frac{4x+21}{x-3}$

$5(x-3)=2(4x+21)$

$5x-15=8x+42$

$-5x \qquad -5x$

$-15=3x+42$

$-42 \qquad -42$

$-57=3x$

$x=-19$

9. $|x|=14$

$x=14 \qquad x=-14$

$x=\{-14,14\}$

10. $|8m-8|=40$

$8m-8=40$ or $8m-8=-40$

$+8 \quad +8 \qquad +8 \quad +8$

$\frac{8m}{8}=\frac{48}{8} \qquad \frac{8m}{8}=\frac{-32}{8}$

$m=6 \qquad m=-4$

$m=\{-4,6\}$

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11. $\left|\frac{m}{7}\right| + 4 = 5$

$-4 \quad -4$

$\left|\frac{m}{7}\right| = 1$

$7 \cdot \frac{m}{7} = 1 \cdot 7 \qquad 7 \cdot \frac{m}{7} = -1 \cdot 7$

$m = 7 \qquad m = -7$

$m = \{-7, 7\}$

12. $\frac{-5|4p+4|}{-5} = \frac{-40}{-5}$

$|4p+4| = 8$

$4p+4 = 8 \qquad 4p+4 = -8$

$4p = 4 \qquad 4p = -12$

$p = 1 \qquad p = -3$

$p = \{-3, 1\}$

13. Solve $a = \frac{v}{t}$ for v. (multiply by t)

$at = v$

14. Solve $-8 = \frac{1}{4}(m-n)$ for m. (multiply by 4)

$-32 = m - n$

$+n \quad +n$

$-32 + n = m$

15. Solve $a^2 + b^2 = c^2$ for b.

$-a^2 \quad -a^2$

$\sqrt{b^2} = \sqrt{c^2 - a^2}$

$b = \pm\sqrt{c^2 - a^2}$

16. Solve $S = 2\pi r(h+r)$ for h. (divide by $2\pi r$)

$\frac{S}{2\pi r} = h + r$

$-r \quad -r$

$\frac{S}{2\pi r} - r = h$

Directions: Use the **3-step process** to solve each word problem.

17. The larger of two numbers is two more than five times the smaller number. If the sum of the two numbers is 80, find both numbers.

let x = smaller #

let $5x+2$ = larger #

$x + 5x + 2 = 80$

$6x + 2 = 80$

$-2 \quad -2$

$\frac{6x}{6} = \frac{78}{6}$

$x = 13$

13, 67

18. Find three consecutive odd integers such that the sum of the smallest number and middle number is 27 less than 3 times the largest number.

let x = 1st consec odd #

let $x+2$ = 2nd consec odd #

let $x+4$ = 3rd consec odd #

$x + x + 2 = 3(x+4) - 27$

$2x + 2 = 3x + 12 - 27$

$2x + 2 = 3x - 15$

$-2x \quad -2x$

$2 = x - 15$

$+15 \quad +15$

$x = 17$

17, 19, 21

19. The length of a flat screen TV is 5 inches less than 3 times its width. If the perimeter of the TV is 150 inches, find the dimensions.

let x = width

let $3x-5$ = length

$2(x) + 2(3x-5) = 150$

$2x + 6x - 10 = 150$

$8x - 10 = 150$

$+10 \quad +10$

$\frac{8x}{8} = \frac{160}{8}$

$x = 20$

20in, 55in

20. Olivia and Camille went clothes shopping. Camille spent \$19.40 less than twice what Olivia spent. In total, they spent \$272.35. Find the amount each girl spent.

let x = Olivia's spending

let $2x - 19.40$ = Camille's spending

$x + 2x - 19.40 = 272.35$

$3x - 19.40 = 272.35$

$+19.40 \quad +19.40$

$\frac{3x}{3} = \frac{291.75}{3}$

$x = 97.25$

Olivia spent \$97.25 and Camille spent \$175.10.

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Name: _____ **Unit 2:** Equations & Inequalities

Date: _____ Bell: _____ **Homework 9:** Equations & Word Problems Review

**** This is a 2-page document! ****

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $-88 = -3(4m + 5) - (1 - 3m)$

$-88 = -12m - 15 - 1 + 3m$

$-88 = -9m - 16$

$+16 \quad +16$

$-72 = -9m$

$\frac{-72}{-9} = \frac{-9m}{-9}$

$m = 8$

2. $-4(5k - 7) + 2k = -2(9k + 5)$

$-20k + 28 + 2k = -18k - 10$

$-18k + 28 = -18k - 10$

$+18k \quad +18k$

$28 \neq -10$

$\emptyset$

3. $7x + 31 = 8x - \frac{1}{3}(27x + 3)$

$7x + 31 = 8x - 9x - 1$

$7x + 31 = -x - 1$

$+x \quad +x$

$8x + 31 = -1$

$-31 \quad -31$

$8x = -32$

$x = -4$

4. $14w - 2(1 - w) = 2(5w - 1)$

$14w - 2 + 2w = 10w - 2$

$16w - 2 = 10w - 2$

$-10w \quad -10w$

$6w - 2 = -2$

$+2 \quad +2$

$6w = 0$

$w = 0$

5. $\frac{4}{5} = \frac{12}{2y - 3}$

$60 = 4(2y - 3)$

$60 = 8y - 12$

$+12 \quad +12$

$\frac{72}{8} = \frac{8y}{8}$

$y = 9$

6. $\frac{6a + 7}{4} = \frac{a - 1}{2}$

$2(6a + 7) = 4(a - 1)$

$12a + 14 = 4a - 4$

$-4a \quad -4a$

$8a + 14 = -4$

$-14 \quad -14$

$8a = -18$

$a = \frac{-9}{4}$

7. $|r + 9| = 4$

$r + 9 = 4$ or $r + 9 = -4$

$-9 \quad -9 \qquad -9 \quad -9$

$r = -5 \qquad r = -13$

$r = \{-13, -5\}$

8. $|2v - 1| = 17$

$2v - 1 = 17$ or $2v - 1 = -17$

$+1 \quad +1 \qquad +1 \quad +1$

$\frac{2v}{2} = \frac{18}{2} \qquad \frac{2v}{2} = \frac{-16}{2}$

$v = 9 \qquad v = -8$

$v = \{-8, 9\}$

9. $\frac{|3 - 6p|}{-5} \cdot -5 = -9 \cdot -5$

$|3 - 6p| = 45$

$3 - 6p = 45$ or $3 - 6p = -45$

$-3 \quad -3 \qquad -3 \quad -3$

$-6p = 42 \qquad -6p = -48$

$p = -7 \qquad p = 8$

$p = \{-7, 8\}$

10. $2|g - 5| + 7 = 13$

$-7 \quad -7$

$\frac{2|g - 5|}{2} = \frac{6}{2}$

$|g - 5| = 3$

$g - 5 = 3$ or $g - 5 = -3$

$+5 \quad +5 \qquad +5 \quad +5$

$g = 8 \qquad g = 2$

$g = \{2, 8\}$

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11. Solve $c = \frac{1}{2}ab^2$ for b.

$\frac{2c}{a} = \frac{ab^2}{a}$

$\sqrt{\frac{2c}{a}} = \sqrt{b^2}$

$b = \pm\sqrt{\frac{2c}{a}}$

12. Solve $\frac{x+8y}{4} = 9y$ for x

$x + 8y = 36y$

$-8y \quad -8y$

$x = 28y$

Directions: Use the **3-step process** to solve each word problem.

13. The larger of two numbers is nine less than four times the smaller number. If the sum of the two numbers is 76, find both numbers.

let x = smaller #

let $4x-9$ = larger #

$x + 4x - 9 = 76$

$5x - 9 = 76$

$+9 \quad +9$

$\frac{5x}{5} = \frac{85}{5}$

$x = 17$

17, 59

14. Find two consecutive odd numbers such that the sum of the larger number and twice the smaller number is 27 less than four times the smaller number.

let x = 1st consec odd #

let $x+2$ = 2nd consec odd #

$x + 2 + 2(x) = 4(x) - 27$

$x + 2 + 2x = 4x - 27$

$3x + 2 = 4x - 27$

$-3x \quad -3x$

$2 = x - 27$

$+27 \quad +27$

$29 = x$

29, 31

15. The length of a rectangle is two more than three times its width. If the perimeter of the rectangle is 92, find the dimensions.

let x = width

let $3x+2$ = length

$2(x) + 2(3x+2) = 92$

$2x + 6x + 4 = 92$

$8x + 4 = 92$

$-4 \quad -4$

$\frac{8x}{8} = \frac{88}{8}$

$x = 11$

11 m, 35 m

16. Maria and Tim sold candy bars for a fundraiser. Maria sold seven less than twice the number of candy bars that Tim sold. In total, Maria and Tim sold 137 candy bars. How many candy bars did Maria sell?

let x = Tim's sales

let $2x-7$ = Maria's sales

$x + 2x - 7 = 137$

$3x - 7 = 137$

$+7 \quad +7$

$\frac{3x}{3} = \frac{144}{3}$

$x = 48$

89 candy bars

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